UNIT I: Vehicle Structure & Engine Components

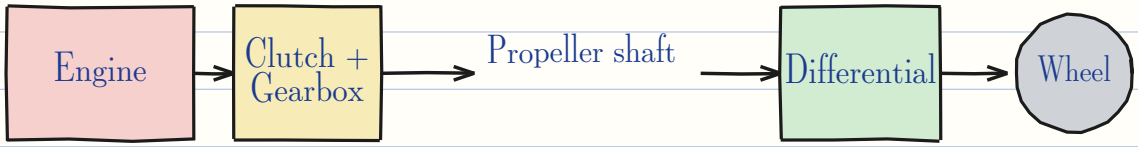
Syllabus: Vehicle construction, chassis & body, specifications, engine types, location, cylinder arrangement, block, head, liners, piston, rings, pin, connecting rod, crankshaft, valves, lubrication, oil pumps, filters, crankcase ventilation.

1. Vehicle Construction

- An automobile is a self-propelled road vehicle that carries passengers or goods, powered by its own engine.
- Basic parts of any vehicle: Chassis + Body (together they make the complete vehicle).
- Chassis = whole vehicle except the body, i.e. everything needed to make it run (a "running chassis").

Main components of the chassis

- Frame - the backbone that carries all the loads.
- Engine - power source (petrol / diesel / hybrid / electric motor).
- Transmission - clutch, gearbox, propeller shaft, final drive, differential, axles.
- Suspension - springs + shock absorbers; isolates road shocks.
- Steering system - controls direction of travel.
- Brakes - slow down / stop / park the vehicle.
- Wheels and tyres - support weight, give traction.
- Electrical system - battery, alternator, starter, lights, ignition.



Power flow in a front-engine rear-wheel-drive (FR) chassis

Fig 1.1 Power flow: engine -> wheels

Remember

Complete vehicle = chassis + body. Chassis without body is called a "chassis cab" or rolling chassis; used for trucks and buses where body is built by a body-builder.

2. Chassis, Body & Specifications

Frame (chassis frame) - functions

- Supports the engine, transmission, body and passengers; carries static + dynamic loads.
- Resists torsion, bending (vertical loads) and side thrust / braking forces.

Types of frame / construction

1. Conventional (ladder) frame - two long side members + cross members. Used in trucks, buses, jeeps. Body is bolted on separately; heavy but strong.
2. Integral / unitary (monocoque) frame - no separate frame; body panels + floor pan form one stressed structure. Used in cars. Light, low floor, less cost; harder to repair.
3. Semi-integral frame - partial frame (sub-frame) at front carries engine and suspension, rest is integral body.

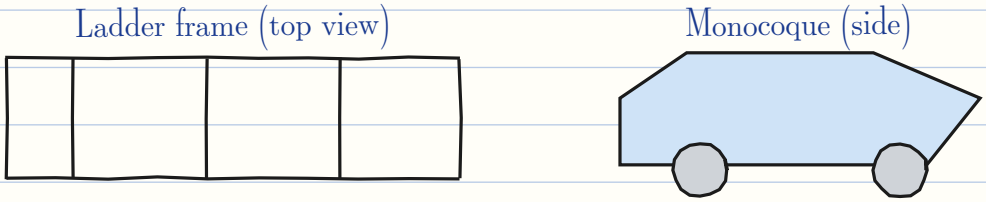


Fig 1.2 Ladder frame and unitary body

Body - functions & types

- Provides passenger / luggage space, protection, aerodynamics and appearance.
- Car body types: saloon (sedan), hatchback, estate, coupe, convertible, SUV, MPV.
- Commercial: truck (platform, tipper, tanker), bus (single / double decker), van.

Vehicle specifications (typical data sheet)

Item	Typical entry
Engine	Type, no. of cylinders, bore x stroke, capacity (cc), CR
Power / Torque	kW (bhp) @ rpm ; Nm @ rpm
Transmission	5-speed manual / AMT / CVT / AT
Dimensions	Length, width, height, wheelbase, track, ground clearance
Weights / others	Kerb weight, GVW, tyre size, fuel tank, max speed

3. Engine - Types

Engine = a machine that converts heat energy of fuel into mechanical work. In vehicles we mainly use Internal Combustion (IC) engines - combustion takes place inside the cylinder.

Classification of engines

Basis	Types
Type of combustion	IC engine, EC engine (steam engine, Stirling)
Fuel used	Petrol (SI), Diesel (CI), Gas / CNG / LPG, bi-fuel, ethanol
Ignition	Spark Ignition (SI) , Compression Ignition (CI)
Cycle of operation	Otto (SI), Diesel (CI), Dual
No. of strokes	2-stroke, 4-stroke
Cooling	Air-cooled, water (liquid) cooled
Cylinder arrangement	In-line, V, flat (boxer), radial, W
Valve arrangement	L, I, F, T head - OHV, OHC (SOHC / DOHC)

Valve arrangements (head types)

- L-head (side valve) - both valves on one side of the cylinder; old engines.
- I-head (overhead valve, OHV) - valves in cylinder head, operated by push rods & rockers.
- OHC - camshaft in the head: SOHC (one cam) or DOHC (two cams, 4 valves/cyl). Best breathing at high rpm.

Terms used in engines

TDC / BDC = top / bottom dead centre. Stroke L = distance TDC to BDC.
Bore D = cylinder diameter.
Swept volume Vs = (pi/4) D^2 L. Compression ratio r = (Vs + Vc) / Vc (Vc = clearance volume).

4. Four-Stroke Engine Working

One cycle = 4 strokes = 2 crankshaft revolutions (720 deg). Only 1 power stroke per 2 revolutions. Valves: inlet (IV) and exhaust (EV).

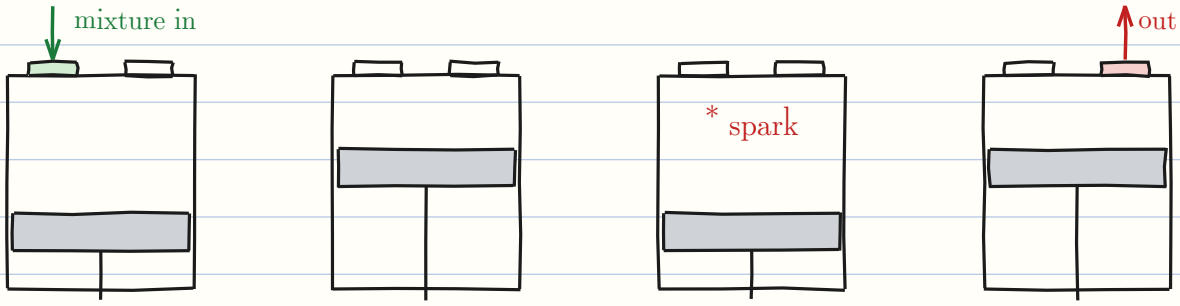


Fig 1.3 Four strokes of the Otto cycle (SI engine)

Strokes

1. Suction: piston TDC -> BDC; IV open, EV closed; fresh air-fuel mixture (petrol) / only air (diesel) is drawn in.
2. Compression: BDC -> TDC; both valves closed; charge compressed (CR 7-11 petrol, 16-22 diesel); temperature and pressure rise.
3. Power (expansion): spark (petrol) / fuel injection & self-ignition (diesel) near TDC; burnt gases push piston TDC -> BDC; work is produced.
4. Exhaust: BDC -> TDC; EV open, IV closed; burnt gases pushed out.

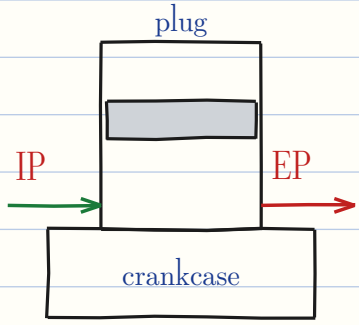
- Flywheel stores energy of power stroke to carry the engine through the other 3 (idle) strokes.

Petrol vs Diesel (4-stroke)

Petrol: mixture drawn, spark plug ignites, CR 6-10, Otto cycle. Diesel: only air drawn, fuel injected at end of compression, self-ignition, CR 16-22, Diesel cycle. Diesel engines give higher thermal efficiency (~35-40 %).

5. Two-Stroke Engine & Comparison

- Two-stroke engine completes a cycle in 2 strokes = 1 revolution of crankshaft (360 deg). Power stroke every revolution.
- No valves - ports (inlet, transfer, exhaust) are opened and closed by the piston itself.
- Crankcase-scavenged: fresh charge is compressed in crankcase and transferred into cylinder.



Upward stroke: compression above piston, suction (fresh charge entering) below piston.

Downward stroke: power above piston, crankcase compression, then transfer port opens; exhaust port opens first.

IP = inlet port, EP = exhaust port (TP = transfer port)

Fig 1.4 Crankcase scavenged 2-stroke petrol engine

Comparison: 2-stroke vs 4-stroke

Point	2-stroke	4-stroke
Power stroke	Once per revolution	Once per 2 revolutions
Valves	Ports (no valves)	Inlet + exhaust valves
Power for same size	Higher (theoretically 2x)	Lower
Fuel economy / emissions	Poor, more HC (oil mixed)	Better, cleaner
Lubrication	Petroil (oil mixed in fuel)	Separate oil sump
Use	Scooters (old), chainsaw, outboard	Cars, trucks, bikes

Scavenging

Process of pushing out burnt gases with fresh charge. Types: cross-flow, loop (Schnurle), uniflow. Loss of fresh mixture with exhaust (short-circuiting) is a drawback of 2-stroke petrol engines.

6. Location of Engine & Drive Layouts

The position of the engine and driven wheels decides interior space, weight distribution, traction and handling.

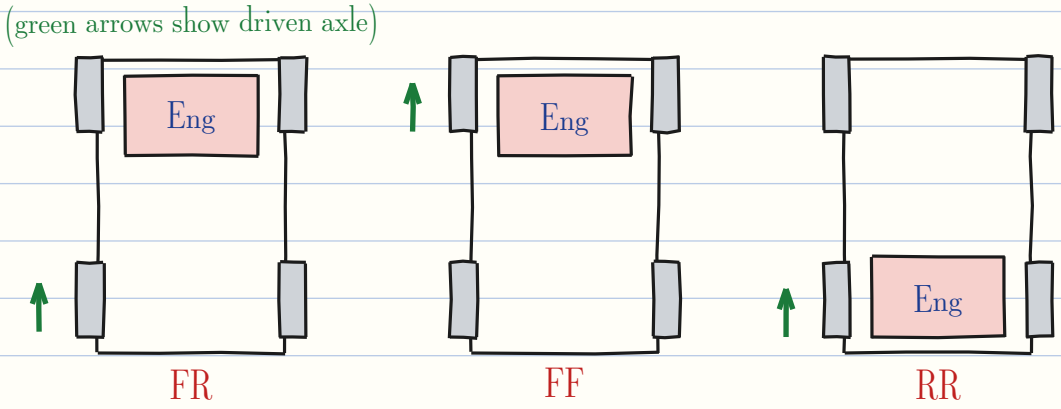


Fig 1.5 FR, FF and RR layouts (top view)

Layouts

- Front engine - rear wheel drive (FR): conventional. Needs propeller shaft; good weight distribution, better traction on hills; used in trucks, SUVs, sports cars.
- Front engine - front wheel drive (FF): compact, no propeller shaft or tunnel, more cabin space, good stability; most modern small cars. Disadvantage: understeer, complex CV joints.
- Rear engine - rear wheel drive (RR): engine over drive wheels -> good traction; used in Tata Nano, VW Beetle, Porsche 911, some buses. Disadvantage: oversteer, cooling problems.
- Mid-engine (MR): engine in between axles, best balance; sports/race cars (Ferrari, Lamborghini).
- All-wheel / 4-wheel drive (AWD/4WD): power to all wheels via transfer case - off-roaders.

Advantage of front engine

Better engine cooling (air flow), simple controls, large luggage space, easy service.

7. Cylinder Arrangement

Number and arrangement of cylinders affects balance, smoothness, size and power. Multi-cylinder engines give smoother torque than a single cylinder.

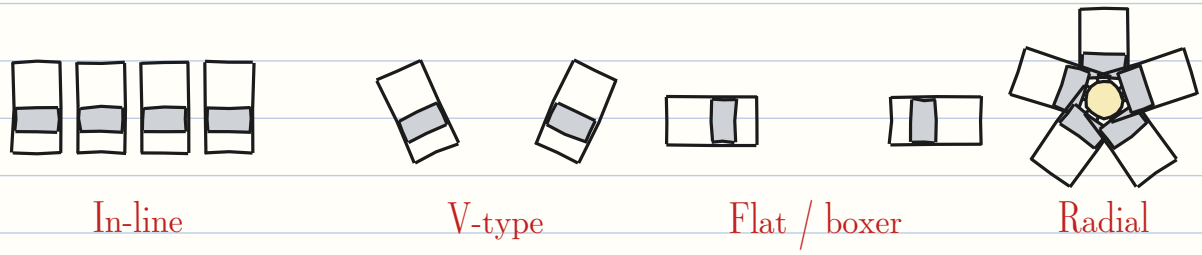


Fig 1.6 Common cylinder arrangements

- In-line: all cylinders in a single row; 3, 4, 5, 6 cyl.; simple & cheap, long engine. Inline-4 most common.
- V-type: two banks at 60-90 deg; short and rigid; V6, V8, V12 - used in luxury / high power cars.
- Flat / horizontally-opposed (boxer): low centre of gravity, perfectly balanced; Subaru, Porsche.
- Radial: cylinders around a crankcase like a star; aircraft engines.
- W-type: three banks (VW W12 / Bugatti W16); Slant engines tilted to lower the bonnet.

Firing order

- Sequence in which power strokes occur in cylinders. Chosen to get even torque, good balance, minimum vibration, uniform cooling.

Engine	Firing order
4-cylinder in-line	1 - 3 - 4 - 2 or 1 - 2 - 4 - 3
6-cylinder in-line	1 - 5 - 3 - 6 - 2 - 4
V8	1 - 8 - 4 - 3 - 6 - 5 - 7 - 2
3-cylinder	1 - 2 - 3

- Firing interval for a 4-stroke = 720 / number of cylinders (4 cyl -> 180 deg; 6 cyl -> 120 deg).

8. Cylinder Block & Crankcase

- The cylinder block is the main structure of the engine - it holds cylinders, water jackets, oil galleries and supports crankshaft, head and accessories.
- Upper part contains the cylinders (bores); lower part is the crankcase which houses crankshaft & main bearings; the oil pan (sump) is bolted below it.

Materials

- Grey cast iron (with Ni, Cr) - good wear resistance, damping, low cost; heavier.
- Aluminium alloy - light (up to 50 % less), good heat transfer; needs cast-iron liners or special bore coating (Nikasil).

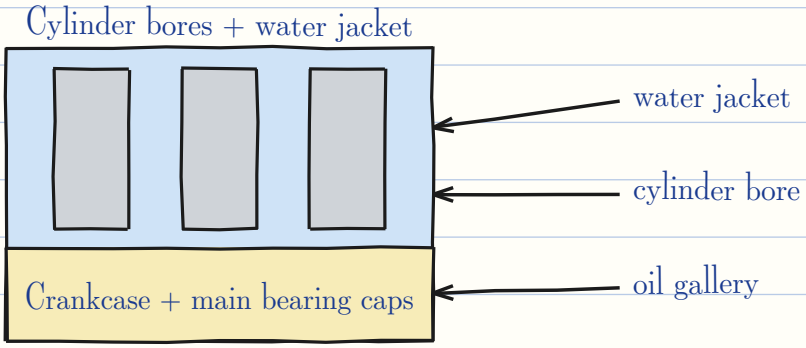


Fig 1.7 Sectional view of a cylinder block

Features of the block

- Water jackets around cylinders (liquid cooling) or fins (air cooling).
- Oil galleries / drilled passages to feed crank bearings, cam bearings and head.
- Deck surface machined flat for head gasket; tappet bores, motor mounting lugs.
- Open deck (jacket open at top, easy casting) vs closed deck (stronger; better for turbo).
- Cylinder bores may be cast in block ("integral") or fitted with liners.

Requirements

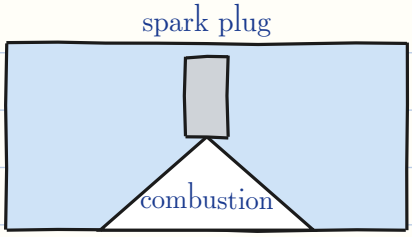
High strength & rigidity, good thermal conductivity, minimum distortion, easy casting & machining, resistance to wear and corrosion.

9. Cylinder Head & Head Gasket

- Cylinder head is bolted on top of the block; it closes the cylinders and forms the combustion chamber.
- Carries inlet & exhaust valves, ports, spark plug (SI) / injector (CI), valve seats, guides, rocker arms or camshafts, water & oil passages.
- Material: cast iron (old) or aluminium alloy (modern - lighter, cooler running).

Combustion chamber shapes (SI)

- Wedge, hemispherical (crossflow, spark plug central - best), bath-tub, pent-roof (4 valve), Heron / bowl-in-piston.
- Good chamber = high compression ratio without knock, short flame travel, turbulence, compact (low surface/volume ratio).



Hemispherical chamber
- valves inclined, large valve area
- central plug, short flame path

Fig 1.8 Hemispherical combustion chamber

Head gasket

- Thin sheet placed between block and head to seal compression gases, coolant and oil.
- Types: copper-asbestos (old), steel-asbestos, multi-layer steel (MLS), composite graphite.
- Head bolts are tightened in a specified sequence (spiral from centre outwards) to the correct torque, in 2-3 stages.
- Blown head gasket symptoms: white smoke, overheating, coolant in oil (milky), loss of compression.

Diesel head

Contains pre-combustion chamber / swirl chamber (IDI) or injector directly over piston bowl (DI); glow plug for cold starting.

10. Cylinder Liners

- A liner (sleeve) is a cylindrical lining inserted into the block; the piston rings slide on its inner surface.
- Why use liners? (1) Block can be made of light material, (2) liner can be hard wear-resistant, (3) worn liner is replaced instead of the whole block, (4) easy repair / reboring.
- Material: alloy cast iron (Ni, Cr, Mo), nitrided steel; inner surface honed (cross-hatch pattern) to hold oil film.

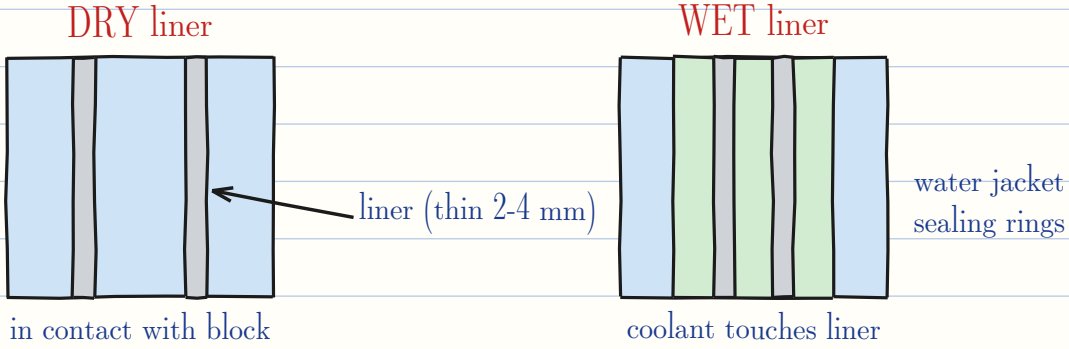


Fig 1.9 Dry and wet cylinder liners

	Dry liner	Wet liner
Contact with coolant	No (pressed into block bore)	Yes - outer surface is cooled directly
Thickness	Thin (1.5 - 4 mm)	Thick (5 - 8 mm)
Cooling	Poorer	Better
Sealing	No sealing problem	Needs rubber rings / flange to stop leakage
Replacement	Needs press / machine	Easy to remove & fit
Use	Light-duty petrol engines	Heavy-duty diesel engines

- Honing, chrome plating, nitriding of liner bore improve wear life. Wear is greatest near TDC (top ring reversal) -> ridge forms.
- Aluminium blocks use cast-in liners or Nikasil / plasma-sprayed coatings.

11. Piston

- The piston moves in the cylinder; it receives gas force and transmits it to the connecting rod through the piston pin; seals the cylinder with rings.

Piston parts

- Crown / head - top surface; flat, domed, or with bowl (diesel).
- Ring belt / lands & grooves - hold compression rings and oil control ring.
- Skirt - guides piston, takes side thrust. Boss - holds gudgeon pin. Thrust faces - minor / major thrust sides.

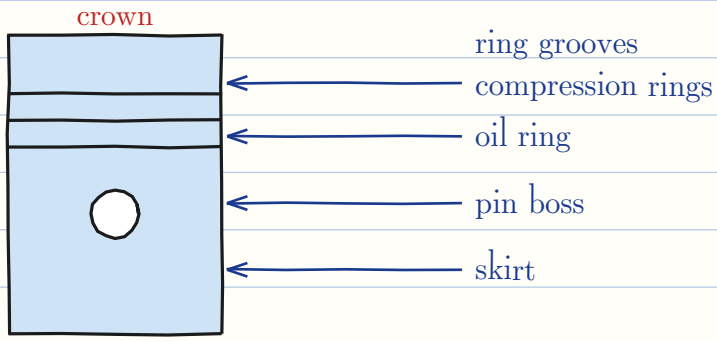


Fig 1.10 Piston (sectional side view)

Materials & design

- Aluminium alloy (Al-Si, 12 % Si "Y-alloy / Lo-Ex") - light, high conductivity; cast or forged. Cast iron pistons only in slow diesels.
- Aluminium expands more than cast iron -> running clearance 0.03 - 0.1 mm; tapered / elliptical (cam-ground) skirt, crown narrower as it is hotter.
- Slots (T-slot, U-slot) give flexibility and heat barrier; steel struts (autothermic) control expansion.
- Skirt types: slipper skirt (cut-outs, light), split skirt, solid skirt, offset pin to reduce piston slap.

Piston slap

Knocking noise from excess clearance when piston changes side-thrust direction at TDC - reduced by offsetting the pin, tapered skirt, correct clearance.

12. Piston Rings & Piston Pin

Piston rings

- Split rings fitted in grooves; spring outward to press against cylinder wall.
- Functions: (1) seal combustion gas (compression), (2) control oil film (scraping), (3) transfer heat from piston to the cylinder wall.

Types

- Compression rings (top 2) - barrel-faced / taper-faced, chrome plated; top ring takes highest heat & wear.
- Oil control (scraper) ring (bottom) - slotted, with expander spring; returns excess oil to sump through holes in groove.

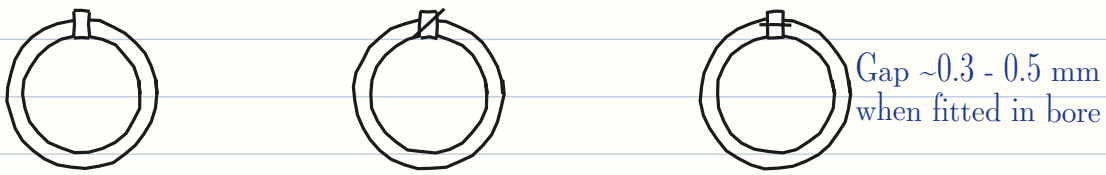


Fig 1.11 End-gap (joint) types

- Ring materials: grey cast iron (alloyed), steel with chrome / molybdenum coating. Rings are staggered 120 deg apart when fitting.
- Faults: ring sticking (carbon), ring wear -> blow-by, high oil consumption, blue smoke.

Piston pin (gudgeon pin / wrist pin)

- Hollow, case-hardened steel pin connecting piston and small end of connecting rod; carries the gas load.
- Hollow = lighter, reduces inertia, strong in bending & shear; ground and polished surface.

Type	Arrangement
Fixed (in piston)	Pin clamped by bolt in boss; free to turn in rod small-end bush
Semi-floating	Pin fixed to rod (clamp), free to turn in piston bosses
Full-floating	Free in both piston & rod; held by circlips at ends - most common

13. Connecting Rod

- Links piston and crankshaft; converts the reciprocating motion of piston into rotary motion of crank and transmits the force.
- Parts: small end (piston pin, bronze bush), shank (I-section), big end (split, two bearing shells with bolts & cap).

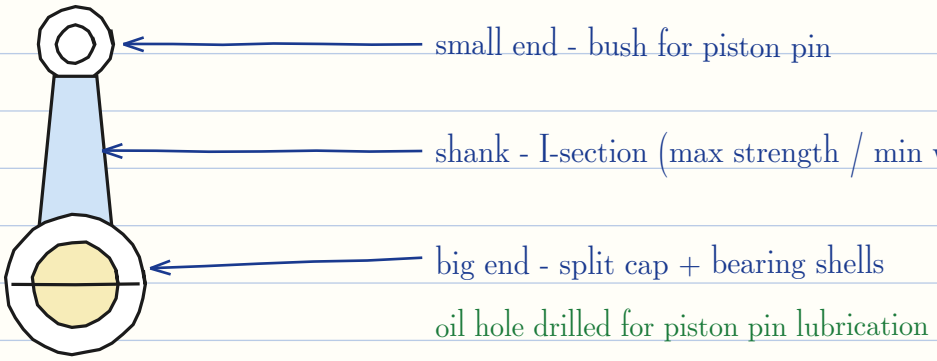


Fig 1.12 Connecting rod

Materials

- Carbon steel (0.4 %C), nickel / chromium-molybdenum alloy steel forged (drop-forged), aluminium (small / racing), titanium (racing), powder metal / cracked-cap rods.

Design points

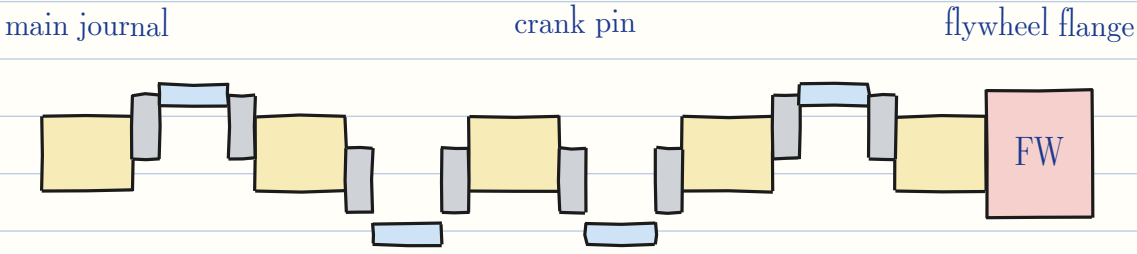
- I-section best: high bending strength, min. weight (small inertia); H-section also used in racing engines.
- Length of rod / crank radius ratio (l/r = 3.5 - 4.5) - longer rod reduces side thrust on piston but increases engine height.
- Big end bolts under high tension; made of high tensile steel and self-locking nuts.
- Bearing shell: steel backing + lining of babbitt / copper-lead / aluminium-tin, with locating tang; thickness uniform.
- Rods are weight-matched per cylinder to minimize vibration; marks on the cap and rod must match (numbered).

Stresses on connecting rod

Compression (gas pressure), tension (inertia at TDC), bending due to whirling (buckling). Designed as strut using Euler / Rankine formula.

14. Crankshaft

- The crankshaft converts reciprocating motion of pistons into rotary motion and delivers torque to the flywheel & transmission.
- Parts: main journals (run in main bearings), crank pins (big end journals), crank webs / arms, counterweights, fillets, oil holes, flange (flywheel), nose (pulley / timing gear).



webs + counterweights balance the rotating & reciprocating masses

Fig 1.13 Four-cylinder crankshaft (schematic)

Materials & manufacture

- Forged alloy steel (Ni-Cr-Mo / EN 8, EN 24 / 40Cr) - high strength; cast nodular (SG) iron - cheaper, good damping (cars).
- Journals are induction hardened / nitrided, ground and polished; fillets rolled to avoid fatigue cracks.

Features

- Number of main bearings: n+1 (for n cyl) = strongest; 3 for 4-cyl is common (economy).
- Crank throw arrangement: 4-cyl in-line -> 180 deg; 6-cyl in-line -> 120 deg; V8 -> 90 deg (cross-plane).
- Torsional vibration damper (harmonic balancer) on nose end absorbs twisting vibrations.
- Flywheel at the rear stores energy & smooths torque; carries ring gear for starter and clutch face.

Crankshaft failure

Fatigue crack at fillets, bearing seizure (lack of oil), bent shaft; check run-out & journal ovality / taper with micrometer.

15. Valves & Valve Train

- Poppet (mushroom) valves open and close the intake & exhaust ports at proper timing; seal the combustion chamber during compression & power.

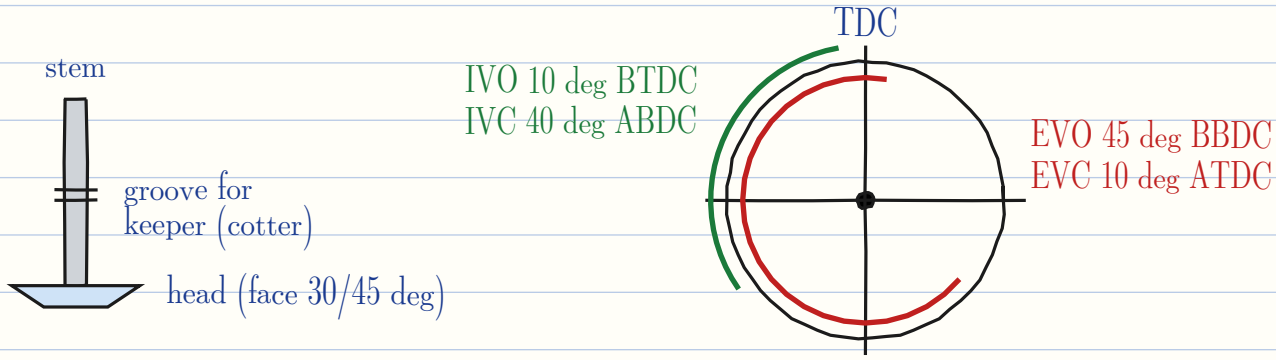


Fig 1.14 Poppet valve and valve timing diagram

Valve materials

- Inlet: chrome / nickel / silicon-chrome steel. Exhaust (runs at 700-800 deg C): austenitic steel (21-4N), Nimonic; stellite-faced seat; sodium-filled hollow stems for cooling.

Parts of valve mechanism

- Camshaft (cam lobes) -> tappet / cam follower -> push rod -> rocker arm -> valve; valve spring (+ retainer & keepers) closes valve; valve seat & valve guide in head.
- Valve clearance (tappet clearance) 0.2-0.3 mm cold; compensates thermal expansion. Hydraulic lifters give zero-lash automatically.
- Camshaft drive: gear, chain or toothed (timing) belt; runs at half crankshaft speed (4-stroke).
- Valve timing: inlet opens before TDC, closes after BDC; exhaust opens before BDC, closes after TDC; overlap period improves breathing. VVT / VTEC alters timing with rpm.
- Faults: valve burning, sticking, spring fatigue, seat recession, wrong tappet gap.

16. Lubrication System

- Lubrication = supplying oil between moving surfaces to form a film that reduces friction & wear.

Functions of lubricating oil

- 1) Reduces friction & wear 2) Cools parts (piston, bearings) 3) Seals piston rings 4) Cleans (carries away carbon, metal particles) 5) Protects against corrosion 6) Damps noise / shock loads.

Types of lubrication systems

1. Petroil / mist system - oil mixed with petrol (2-stroke engines, 20:1 - 50:1).
2. Splash system - dippers on big ends splash oil from troughs; used in small stationary / lawn-mower engines.
3. Pressure feed (full pressure) system - oil pump sends oil under 2-4 bar to all main bearings, big ends, cam bearings; most cars.
4. Semi-pressure - pressure to main bearings, splash for cylinder walls.
5. Dry sump - oil stored in separate tank; scavenge pump returns oil; used in racing / sports cars & aircraft (no oil starvation in corners).
6. Wet sump - oil kept in the sump below crankcase; simple; most vehicles.

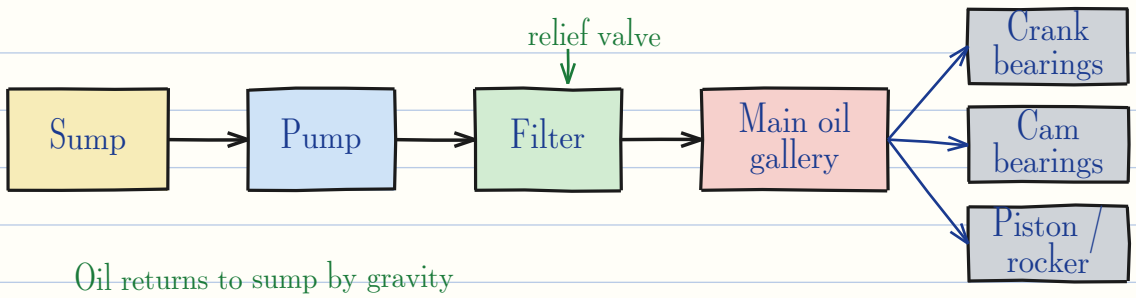


Fig 1.15 Wet-sump pressure feed lubrication

- Oil pressure relief valve keeps pressure constant (~3 bar). Oil pressure warning lamp / gauge on dash.
- Oil grade: SAE viscosity (e.g. 20W-40, 5W-30); API service class SL, SN (petrol), CI-4 (diesel). Viscosity index is high for multigrade oils.

17. Oil Pumps, Filters & Crankcase Ventilation

Oil pumps

- Gear pump: two meshed gears (drive + driven) trap oil between teeth and housing; simple, robust, driven from camshaft / crankshaft.
- Rotor (gerotor) pump: inner rotor with n lobes inside outer rotor with n+1 lobes; compact, quiet, used on the crank nose.
- Vane pump - sliding vanes in eccentric rotor, variable-displacement versions save energy.

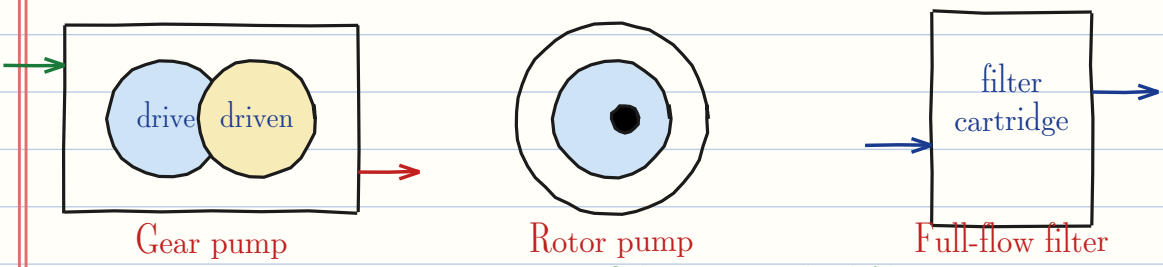


Fig 1.16 Oil pumps and oil filter

Oil filters

- Remove carbon, metal dust, sludge. Element: paper (pleated), felt, cotton waste, metal-mesh. Strainer (wire mesh) on pump pick-up in sump.
- Full-flow filter - all oil passes through; a bypass valve opens if filter clogs. Bypass filter - only 10 % oil filtered continuously (finer). Change every 5000-10000 km.

Crankcase ventilation

- Blow-by gases (unburnt HC, water vapour, acids) leak past rings into the crankcase -> dilute / contaminate oil, cause sludge, corrosion & pressure build-up.
- Open (road-draught) ventilation: breather tube vents fumes to atmosphere - pollution (old).
- Closed / PCV (Positive Crankcase Ventilation): fumes drawn into the intake manifold through the PCV valve and burnt; fresh air enters through the breather / air filter. Reduces HC emission.

PCV valve

Spring-loaded; at idle (high vacuum) it restricts flow, at full load (low vacuum) opens fully. A stuck valve causes rough idle, oil leaks, sludge.

UNIT II: Ignition & Fuel Supply Systems

Syllabus: Ignition system - coil & magneto, spark plug, distributor, electronic ignition; fuel system - carburettor, fuel pumps, fuel injection (mono-point, multi-point), unit injector, nozzle types, EFI, GDI, MPFI, DTS.

1. Ignition System - Introduction

- Ignition system supplies a high-voltage spark (10-30 kV) across the spark plug gap at the correct instant to ignite the compressed air-fuel mixture in SI (petrol) engines.

Functions

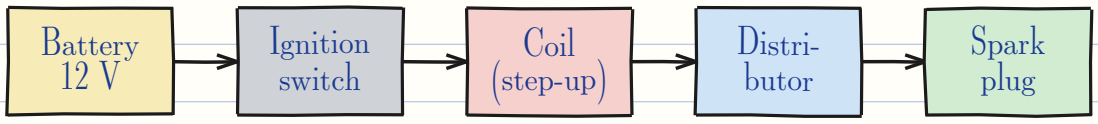
- Step up battery voltage (12 V) to a very high voltage.
- Distribute the spark to each cylinder in the firing order.
- Time the spark correctly (advance with speed & load).

Requirements of a good ignition system

- Strong spark at all speeds & loads (even at cranking).
- Accurate timing; reliable; long life; low maintenance; interference-free (RFI suppression).

Types of ignition systems

1. Battery (coil) ignition - conventional, with contact breaker & distributor.
2. Magneto ignition - self-generating, no battery; two-wheelers, racing, small engines.
3. Electronic ignition - transistorised (TCI), capacitive discharge (CDI), distributorless (DIS), coil-on-plug (COP).



Basic chain: 12 V -> 20,000 V -> spark in cylinder

Fig 2.1 Block diagram of ignition system

Spark energy

Minimum ignition energy ~0.2 mJ (stoichiometric); rich / cold start needs ~3 mJ. Spark lasts about 1-2 ms. Required voltage rises with gap, compression, rpm & lean mixture.

2. Battery (Coil) Ignition System

- Used in cars (older) with contact breaker + distributor. Two circuits: primary (low tension, LT) and secondary (high tension, HT).

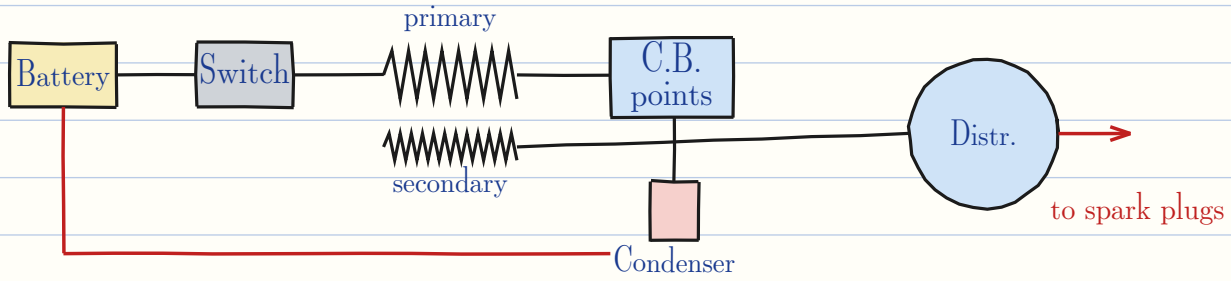


Fig 2.2 Battery coil ignition circuit

Components

- Battery (12 V), ignition switch, ballast resistor (reduces primary voltage at high speed; bypassed while cranking).
- Ignition coil (autotransformer), contact breaker (CB) points + cam, condenser (0.2 uF), distributor, spark plugs, HT leads.

Working

1. Switch ON, CB points closed: current ~3-4 A flows in the primary winding and builds a magnetic field around the iron core.
2. Cam opens the points: primary current is cut suddenly -> field collapses, cutting the many turns of the secondary winding.
3. EMF induced in secondary = 20-30 kV (turns ratio ~ 100:1) -> sent through the distributor rotor to the correct plug -> spark jumps the gap.
4. Condenser absorbs self-induced EMF (~250 V) in the primary, prevents arcing at points and makes the field collapse faster -> stronger spark.

Dwell angle

Angle through which the CB points remain closed (about 60 deg for 4-cyl; ~ 60% of cycle). Short dwell -> weak spark at high speed; long dwell -> points burn.

3. Ignition Coil & Contact Breaker

- The ignition coil is a pulse transformer: low turns on primary (200-300 turns of thick wire) + very many turns on secondary (20,000 - 25,000 turns of fine wire) wound on a laminated soft-iron core.

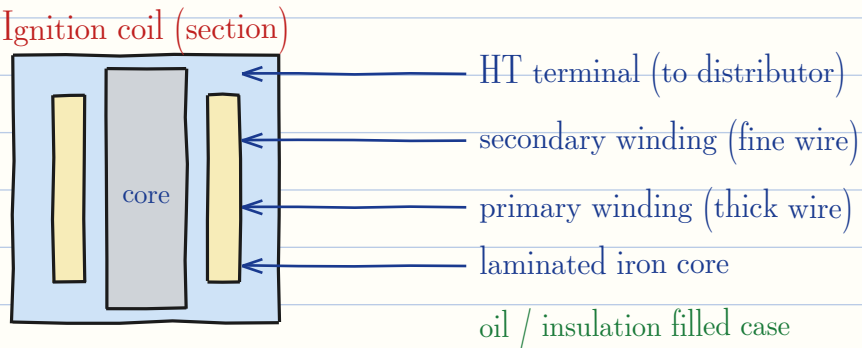


Fig 2.3 Ignition coil construction

Key points

- Turns ratio = N2 / N1 ~ 100:1 ; secondary voltage V2 = V1 x (N2/N1) [theory]; but real voltage depends on rate of flux collapse.
- Energy stored: E = 1/2 L I^2 (joules) - about 30-60 mJ.
- Coil is cased in oil / resin for insulation & cooling. Dry (epoxy) coils are used in modern cars.

Contact breaker (CB) unit

- Spring-loaded moving arm with tungsten contacts, fixed point, fibre rubbing block following the cam (lobes = number of cylinders).
- Point gap 0.35 - 0.45 mm; gap too small -> long dwell, weak spark at low speed; too large -> short dwell.
- Faults: burnt / pitted points, wrong gap, weak spring, faulty condenser (excess arcing), oil on points.

Fault	Effect
Wide point gap	Late timing / short dwell / misfire at high speed
Narrow gap	Early timing / long dwell / burnt points
Faulty condenser	Heavy arcing at points, weak or no spark

4. Magneto Ignition System

- A magneto generates its own current (no battery needed) by rotating magnets near coils - used on two-wheelers, scooters, lawn-mowers, aircraft & racing engines .
- Working principle: same as coil ignition, but the primary current is produced by electromagnetic induction from a rotating permanent magnet.

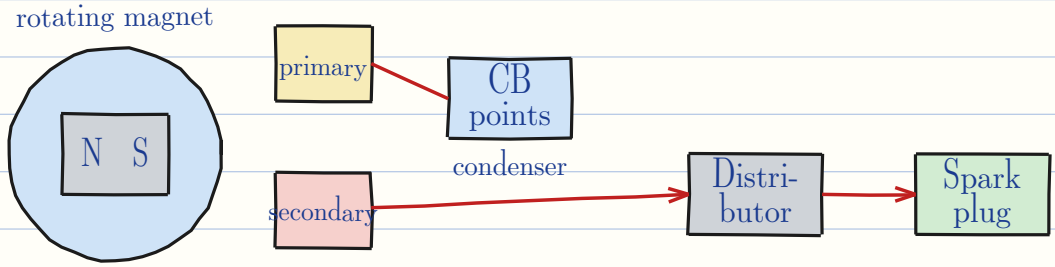


Fig 2.4 Magneto ignition (block view)

Types of magneto

- Armature-rotating type: coils rotate between fixed magnets (old). Magnet-rotating type: magnet rotates, coils fixed - common. Polar-inductor type: only iron cores rotate, magnet & coils fixed.
- Flywheel magneto - magnets in flywheel rim; coil outside; used in scooters, motorcycles (also with CDI).

Coil vs Magneto ignition

Point	Battery coil	Magneto
Battery	Required	Not required
Starting	Good (strong spark at low rpm)	Poor at cranking speed
Spark at high rpm	Weaker	Stronger
Weight / cost	Cheaper, easy repair	Costlier, compact
Use	Cars, buses, LCVs	Two-wheelers, racing, small engines

5. Spark Plug

- The spark plug provides the air gap where the high-voltage spark jumps and ignites the mixture. Screwed into the cylinder head.

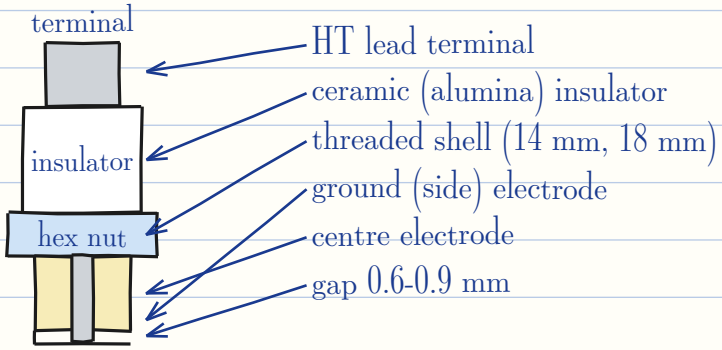


Fig 2.5 Spark plug construction

Parts & materials

- Centre electrode - nickel alloy / copper core / platinum / iridium (long life). Ground electrode welded to shell.
- Insulator - sintered alumina ceramic, withstands 30 kV and 1000 deg C; steel shell with thread; copper gasket seals.

Heat range (hot & cold plugs)

- Heat range = ability to transfer heat from firing tip to the head. Hot plug: long insulator nose, slow heat loss - for low-speed / light-duty engines. Cold plug: short nose, rapid heat loss - for high-speed / high-compression engines.
- Self-cleaning temperature 500 deg C (burns carbon); pre-ignition above 850 deg C.

Condition	Meaning of plug tip
Light brown / grey	Normal; correct heat range
Black, sooty	Rich mixture / plug too cold
White / blistered	Lean / overheating / plug too hot
Oily wet	Oil leaking past rings / valves

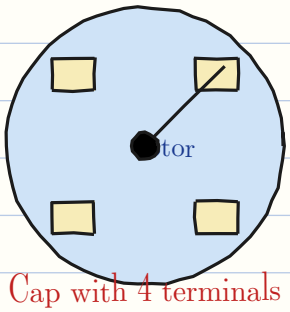
- Gap is set with feeler gauge (0.6 - 0.8 mm); replace every 20,000 - 40,000 km (platinum / iridium ~ 80,000 km).

6. Distributor & Ignition Timing

- The distributor (a) opens/closes primary circuit through CB points, (b) distributes HT to plugs in firing order, (c) advances timing automatically.

Parts

- Cap (bakelite, with central & outer terminals), rotor arm, CB points + cam, condenser, driven shaft (half crank speed), centrifugal advance weights, vacuum advance diaphragm, oil-felt for cam lubrication.



Cap with 4 terminals

Rotor sweeps past terminals in firing order 1-3-4-2

- small gap (0.3 mm) between rotor tip and terminal (spark jumps)

Centre terminal <- HT from coil

Fig 2.6 Distributor cap and rotor (top view)

Automatic spark advance

- Centrifugal advance: bob-weights fly out with speed, rotating cam forward against springs -> spark earlier at high rpm (flame needs more crank angle).
- Vacuum advance: diaphragm connected to inlet manifold vacuum (near throttle) moves CB plate -> spark advanced at part load (lean mixture burns slowly). Retards at full load.
- Octane selector - manual fine adjust of timing for fuel grade.

Ignition timing

Spark occurs 5-15 deg BTDC at idle, up to 30-40 deg BTDC at high speed. Too advanced -> knocking; too retarded -> loss of power, overheating. Checked with a stroboscopic timing light.

7. Electronic Ignition System

- Replaces mechanical CB points with a transistor / thyristor switch triggered by a sensor - no wear, no maintenance, stronger spark (up to 35-40 kV), better cold start & emission.

Types

- TCI (transistorised coil ignition) - pickup (inductive or Hall-effect sensor, or optical) sends pulse to an igniter module which switches the primary current.
- CDI (capacitor discharge ignition) - a capacitor is charged to 300-400 V by a DC-DC converter, then dumped through the coil primary by a thyristor (SCR) -> very fast rise time; used in motorcycles.
- DIS (distributorless / waste-spark) - one coil for two cylinders; both plugs fire together (one in compression, one in exhaust). No rotor or cap.
- COP (coil-on-plug) - individual coil mounted on each plug; no HT leads; ECU controls timing for each cylinder. Best energy; used in modern engines.

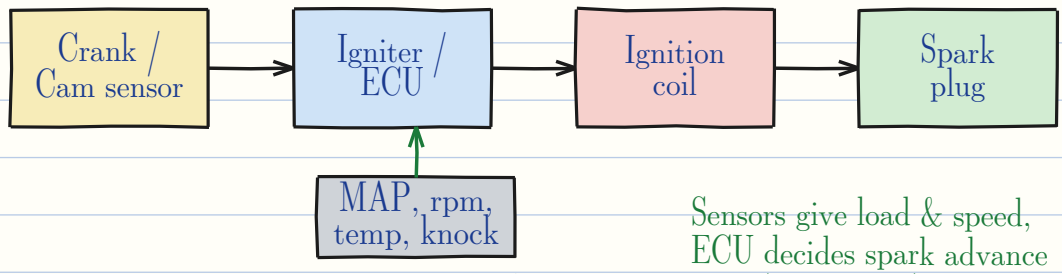


Fig 2.7 Electronic ignition (with ECU)

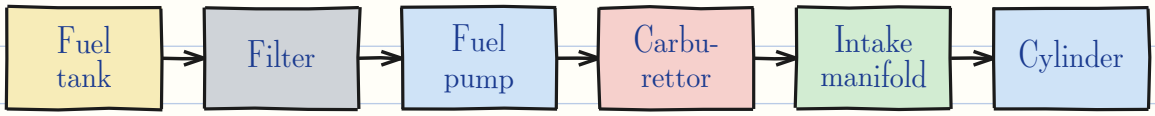
Advantages

- No contact wear / dwell setting; consistent high voltage at all speeds; programmed timing map (3-D map of load x rpm) instead of mechanical advance weights.
- Knock sensor retards timing automatically; allows higher compression ratio; lower emissions and better fuel economy.
- Hall-effect sensor: a rotating shutter (vane) cuts a magnetic field in a Hall IC producing a clean square-wave pulse.

8. Fuel System - Overview

- Purpose: store fuel, filter it, and deliver the correct quantity of fuel (in correct air-fuel ratio) to the engine at all speeds and loads.

Petrol engine - carburettor system



Air cleaner -> air also enters the carburettor
Return line brings excess fuel back to the tank (injection systems)

Fig 2.8 Petrol fuel system (carburettor type)

- Fuel tank (steel / plastic, baffles, vent, filler cap); fuel lines (steel / nylon); filters (sediment bowl, paper element 10-20 micron); pump; carburettor / injectors; air cleaner (paper, oil-bath).

Diesel fuel system (overview)

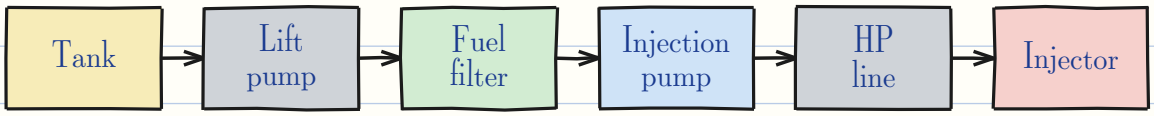


Fig 2.9 Diesel fuel system (inline pump type)

- Diesel = high pressure injection (150-2000 bar) of finely atomised fuel into the compressed air near TDC. Quantity governed by the governor / ECU.

Air-fuel ratio (A/F)

Condition	A/F (by mass)
Idling / cold start	10:1 - 12:1 (rich)
Cruising (economy)	15:1 - 17:1 (lean)
Full power	12:1 - 13:1
Stoichiometric (chemically correct)	14.7 : 1

9. Carburettor - Principle

- A carburettor atomises and vapourises petrol and mixes it with air in the proper ratio for different engine conditions.
- Principle: Bernoulli / venturi effect - air accelerates through a narrow throat (venturi) -> pressure drops (depression 0.05 - 0.07 bar) -> fuel is pushed from the float chamber through the jet into the air stream.

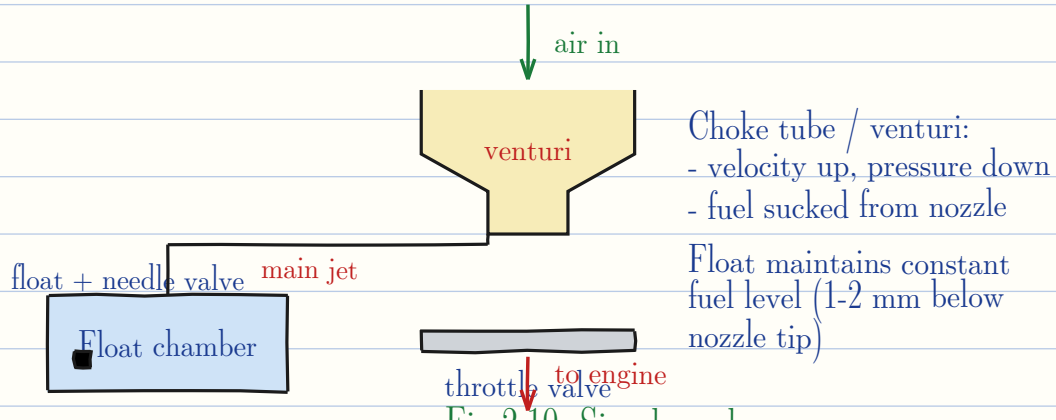


Fig 2.10 Simple carburettor

Parts of a simple carburettor

- Float chamber + float + needle valve (constant level); main jet / metering jet; discharge nozzle; venturi (choke tube); throttle valve (butterfly, controls quantity); choke valve (butterfly at air entrance).
- Defect of simple carburettor: gives increasingly rich mixture as throttle opens (flow rate grows) -> needs compensating devices (air bleed, compensating jet).

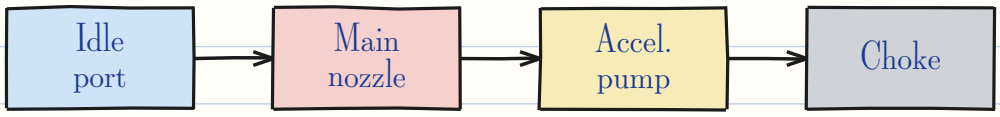
Requirements

Rich mixture for starting & idling, lean for cruising, slightly rich for full power; quick response in acceleration; good atomisation.

10. Carburettor Systems & Types

Practical carburettors (Solex, SU, Zenith, Carter) add circuits so that mixture suits every condition:

Circuit / device	Function
Choke system	Butterfly in air entry; closing it raises depression -> rich mixture for cold starting
Idling system	Idle jet + idle port near throttle edge; supplies rich mixture when throttle is almost closed
Main (cruising) system	Main jet + compensating / air-bleed jet keeps mixture lean (15:1) at medium throttle
Accelerator pump	Piston / diaphragm squirts extra petrol when throttle is opened suddenly (avoids flat spot)
Power enrichment (economiser)	Extra fuel via metering rod / vacuum valve at full load
Anti-percolation / vent	Prevents vapour lock and fuel overflow when hot



Each circuit takes fuel from the same float chamber

Fig 2.11 Circuits of a modern carburettor

Types of carburettors

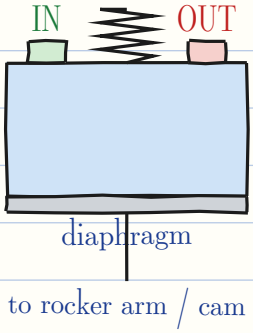
- By draught: updraught (air upward), downdraught (most common - gravity helps), horizontal / side-draught (SU, motorcycles).
- By venturi: fixed venturi (Solex, Zenith) vs constant-depression / variable-venturi (SU, CV - a piston varies the choke area; mixture almost constant).
- Drawbacks: poor atomisation / distribution to cylinders, icing, no accurate A/F control -> replaced by injection (BS-III / BS-IV emission norms).

11. Fuel Pumps

- Pump draws fuel from the tank and delivers it to the carburettor (0.2 - 0.3 bar) or injection rail (3 - 6 bar for MPFI, 100-200 bar for GDI).

1. Mechanical (diaphragm) pump

- Driven by an eccentric on camshaft -> rocker arm -> pulls diaphragm down (suction, inlet valve opens) and spring pushes it up (delivery, outlet valve opens).
- Delivery pressure is limited by the diaphragm spring - when carburettor needle valve closes, the diaphragm stays down (no flow) => self-regulating.



- Suction stroke: diaphragm pulled down, inlet valve opens, fuel enters chamber
- Delivery stroke: spring pushes up, outlet valve opens, fuel to carburettor

Fig 2.12 Mechanical diaphragm fuel pump

2. Electric fuel pump

- Plunger (SU) type - solenoid + contact breaker moves a diaphragm/bellows; mounted near tank. Rotary types: roller-cell, gear (gerotor), turbine (in-tank pump - quiet, cool, prevents vapour lock).
- Used in EFI / MPFI - provides constant high pressure; relay cut-off if engine stops (safety). Delivery 100-150 l/h at 3-5 bar.

3. High-pressure pumps

- GDI: engine-driven piston pump (100-200 bar). Diesel: inline / rotary / common-rail pump (1000-2000 bar). Lift (feed) pump supplies the HP pump.

12. Petrol Fuel Injection - Mono & Multi Point

- Fuel injection = fuel is sprayed under pressure into the intake air by injectors (solenoid valves) controlled by an ECU, instead of being sucked by a carburettor.

Advantages over carburettor

- Accurate A/F ratio for each cylinder; better power, mileage & emissions; quick throttle response; no icing; easy cold start; better cylinder-to-cylinder distribution.

Classification

- Mono-point / throttle-body injection (TBI, SPI): one or two injectors above the throttle valve - low cost, uses existing manifold; mixture still travels through the manifold (wall wetting).
- Multi-point / port injection (MPFI): one injector for each cylinder, near intake valve in the port -> precise & fast. Modes: simultaneous (all fire together), group (banks), sequential (each in firing order, best).
- Direct injection (GDI): injector inside combustion chamber.

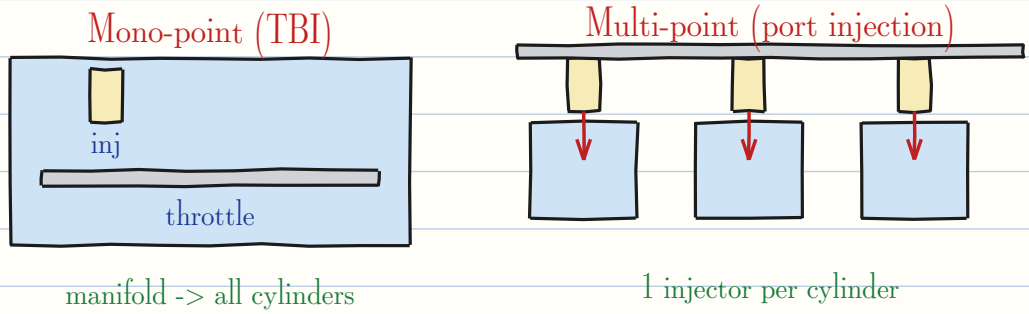


Fig 2.13 Mono-point vs multi-point injection

Point	Mono-point (TBI)	Multi-point (MPFI)
No. of injectors	1 (or 2)	One per cylinder
Mixture control	Fair	Excellent
Cost	Low	Higher
Fuel pressure	~1 bar	3 - 4 bar

13. Diesel Injection & Unit Injector

- Diesel engines need high-pressure, precisely timed and metered injection into the compressed air. Systems:

1. Inline (jerk) pump system - one plunger-barrel element per cylinder driven by a camshaft; quantity controlled by rotating the plunger (helical groove) through a rack; governor controls rack. Heavy trucks.
2. Distributor (rotary) pump - one pumping plunger + distributor rotor serves all cylinders (Bosch VE, CAV DPA). Compact; cars & LCVs.
3. Common rail direct injection (CRDI) - high-pressure pump (1600-2000 bar) fills a common rail; ECU opens solenoid / piezo injectors at any time -> pilot, main & post injection. Quiet, clean, efficient.
4. Unit injector (UI / EUI) and unit pump - pump + injector combined in one unit per cylinder, driven by camshaft rocker; no high-pressure lines (up to 2000-2200 bar).

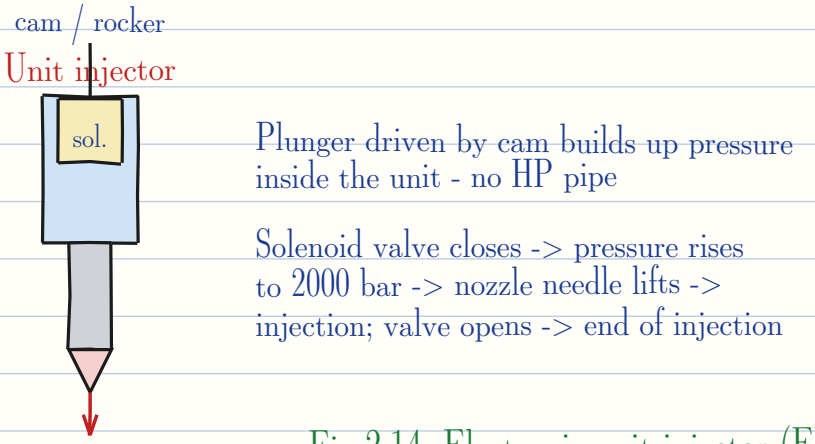


Fig 2.14 Electronic unit injector (EUI)

- Advantages of unit injector: very high pressure, fast response, precise timing by ECU, no pipe pressure waves. Used in Cummins, Caterpillar, Volvo, Detroit heavy-duty engines.
- Common-rail parts: HP pump, rail with pressure sensor & regulator, injectors, ECU, sensors.

14. Injector Nozzle Types

- The nozzle (injector) atomises fuel and distributes it in the combustion chamber. Needle valve is held on its seat by spring; opens when fuel pressure exceeds opening (popping) pressure: 120 - 250 bar (IDI: 100-150; DI: 200-300).
- Holder carries nozzle body, needle, spring, adjusting screw, leak-off (back-leak) line.

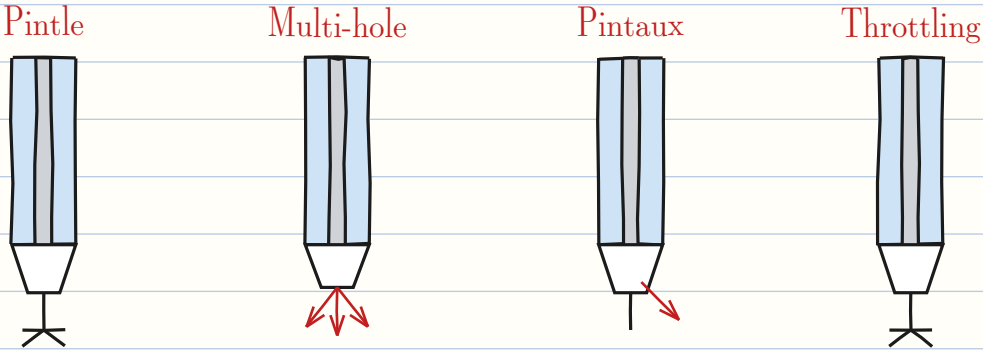


Fig 2.15 Types of diesel injector nozzles

Nozzle	Features / use
Single-hole	One central hole, 8-10 mm spray angle small; 175 bar; open chamber (DI) older engines
Multi-hole	4-8 holes (0.25-0.4 mm) at spray angle ~150 deg; direct injection engines; 200-300 bar
Pintle	Pin protrudes through orifice; spray cone 0-60 deg; self-cleaning; IDI / swirl chambers; 100-135 bar
Delay (throttling) pintle	Pin shaped to give small initial flow -> reduces diesel knock (IDI)
Pintaux	Pintle with auxiliary hole -> cold-start spray on to hot area; 3-hole variation

- Faults: nozzle choking (carbon), needle sticking, dripping, wrong opening pressure -> smoke, knock; tested on a nozzle tester.

15. Electronic Fuel Injection (EFI)

- EFI uses an ECU (electronic control unit) that reads sensors, calculates the required fuel quantity, and opens injectors for a calculated pulse width (1.5-10 ms) - closed-loop control with the lambda (O2) sensor.

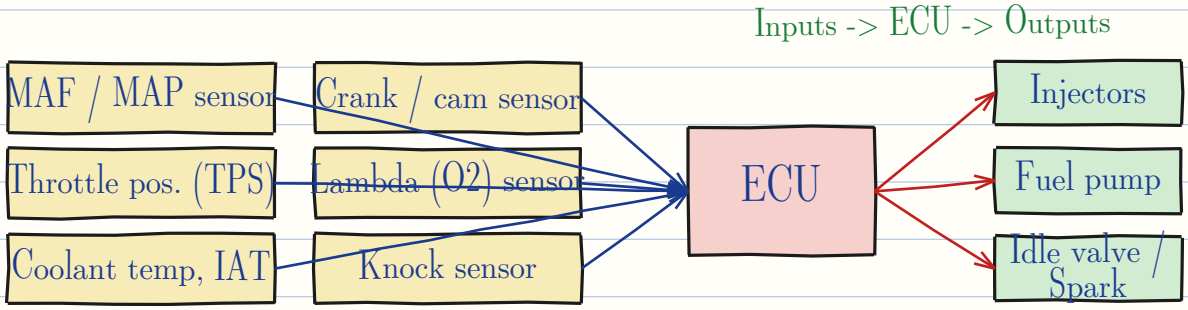


Fig 2.16 EFI system layout (block diagram)

Main sensors

- MAF (hot-wire air-mass), MAP (manifold pressure), TPS (throttle potentiometer), CTS / ECT (NTC thermistor), IAT, crank position (CKP) & cam (CMP), O2 (lambda), knock (piezo).

Actuators

- Solenoid injectors (12-16 ohm), fuel pump, idle air control / electronic throttle, ignition coils, EGR valve, canister purge valve.

Operating principle

- Basic pulse width = f(air mass, rpm) -> corrected for temperature, battery voltage, lambda feedback; modes: start, warm-up, idle, acceleration, full load enrichment, deceleration fuel cut-off.
- Closed loop keeps lambda = 1 so that the three-way catalytic converter works at best efficiency; OBD stores fault codes (DTC).

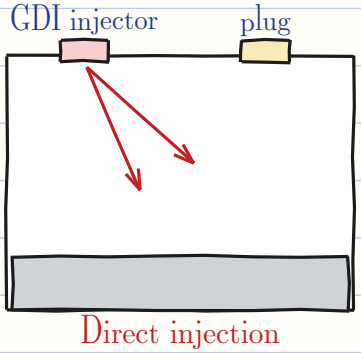
16. MPFI and GDI

MPFI (Multi-Point Fuel Injection)

- Each cylinder has its own injector in the intake port, spraying fuel toward the back of the intake valve; fuel rail at 2.5 - 4 bar.
- Sequential timing: injector fires once per cycle, during the intake stroke of that cylinder.
- Benefits: smooth idle, better mileage (~10 %), lower CO / HC, good cold start, fast response.

GDI (Gasoline Direct Injection)

- Petrol is injected directly into the combustion chamber at 50 - 250 bar by a high-pressure injector (like diesel); engine-driven HP pump + rail + ECU.



Modes:

1. Stratified charge (late injection, compression stroke) - ultra-lean
2. Homogeneous (early injection, intake stroke) - at full load
3. Homogeneous lean / stoich + EGR

Fig 2.17 Gasoline direct injection

- Advantages: 15-20 % better fuel economy, higher compression ratio (10-12), cooling effect of fuel evaporation (less knock), more power & torque, supports turbo-downsizing (TSI, EcoBoost, FSI, Skyactiv-G).
- Disadvantages: costly HP pump & injectors, carbon deposits on intake valves (no fuel wash), more particulate (PM) / NOx -> gasoline particulate filter (GPF), noise.

MPFI vs GDI

MPFI: fuel injected in port (3-4 bar), always homogeneous. GDI: fuel injected in cylinder (50-250 bar), stratified or homogeneous; more efficient but costlier.

17. DTS-i (Digital Twin Spark) & Revision

- DTS-i (Digital Twin Spark ignition) is a Bajaj technology (Pulsar) using two spark plugs per cylinder controlled digitally (CDI).
- Why two plugs? In a wide-bore, flat-piston, short-stroke engine one central plug cannot burn the mixture quickly: the flame has to travel far. Two plugs at opposite ends start two flame fronts that meet in the middle.

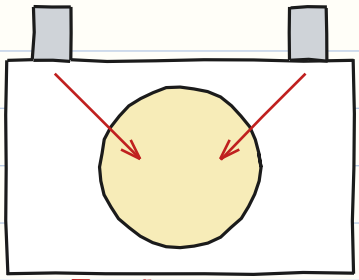


Fig 2.18 Twin spark plugs in combustion chamber

DTS-i family:
- DTS-i : 2 plugs + digital CDI (fast burn)
- DTS-Si : 2 plugs + swirl induction for better air-fuel mixing, higher mileage
- DTS-FI : DTS + electronic fuel injection

Advantages of DTS

- Faster, complete combustion -> more power, better mileage, lower emission; the ECU/CDI can retard / advance each plug independently; smooth running at low speed.
- Plugs may fire together or with a slight lag depending on rpm (map-controlled).

Quick revision - ignition & fuel

System	Key point
Coil ignition	Primary 12 V -> secondary 20-30 kV; CB + condenser
Magneto	Self-generating; two-wheelers
Spark plug	Gap 0.6-0.9 mm; hot / cold by heat range
Carburettor	Venturi principle; A/F 15:1 cruise
MPFI / EFI	ECU + sensors + injector pulse width
GDI	In-cylinder injection 50-250 bar

UNIT III: Steering & Suspension System

Syllabus: Principle of steering, steering geometry & wheel alignment, steering linkages, steering gearboxes, power steering, front axle, suspension - independent & solid axle, coil / leaf / air suspension, torsion bar, shock absorbers.

1. Principle of Steering

- Steering system lets the driver guide the vehicle by turning the front (sometimes all four) wheels. Driver effort at the steering wheel is multiplied by the gearbox and transmitted through linkages to the wheels.

Functions

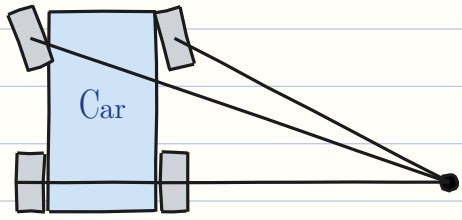
- Control direction of travel; provide directional stability; minimise tyre wear; give driver feel (road feedback); absorb shocks from road.

Requirements

- Easy & light operation, self-centring (return-ability), correct steering geometry, true rolling (no skid) of wheels on a turn.

Condition for true rolling - Ackermann principle

- During a turn all wheels must roll about one common point (instantaneous centre) on the line of the rear axle. Hence the inner wheel must turn more than the outer wheel.
- Condition: cot(phi) - cot(theta) = c / b (c = distance between pivot axes (track), b = wheelbase; theta = outer wheel angle, phi = inner wheel angle).



No skid -> no tyre wear
All wheel axes meet at I
inner wheel angle > outer angle

Fig 3.1 Correct steering (Ackermann condition)

Turning radius

Min. turning circle radius of a car ~ 4.5 - 6 m. Outer-wheel turning radius R = b / sin(theta) (approx.), where b = wheelbase.

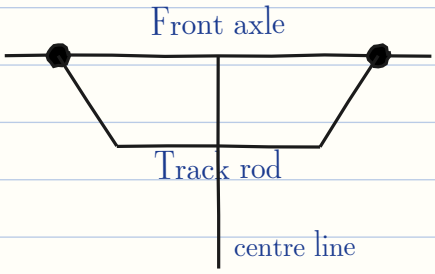
2. Ackermann & Davis Steering Gear

Davis steering gear

- Uses sliding pairs. Gives exact (mathematically correct) steering for all positions but has many parts that wear -> backlash; rarely used.

Ackermann steering gear

- Uses turning pairs only (simple, less wear). The track arms are inclined at angle alpha to the front axle and joined by the track rod. Only approximately correct - exact at straight-ahead and one more position, small error at others - but widely used.



Track arms inclined inwards,
lines through them meet at
a point on the rear axle,
centre of wheel base (approx.)
Used in all modern vehicles

Fig 3.2 Ackermann mechanism (top view)

Point	Davis	Ackermann
Pairs	Sliding (more wear)	Turning (less wear)
Accuracy	Exactly correct	Approximately correct
Maintenance	High	Low
Use	Rare	Universal

Types of steering by wheels

- Front-wheel steering (most vehicles), rear-wheel steering, four-wheel steering (4WS) - rear wheels turn opposite at low speed (tight turns) and same direction at high speed (stability); used in some SUVs / trucks.

Main parts of steering system

- Steering wheel -> steering column (collapsible / tilt) -> steering gearbox -> pitman arm -> drag link / tie rods -> steering arms -> stub axles -> wheels.

3. Steering Geometry - Camber & Caster

Steering geometry (wheel alignment angles) describes the angles of the front wheels & steering axis so that the vehicle is stable, steers easily and the tyres wear evenly. Main angles: camber, caster, king-pin inclination (KPI), toe-in & toe-out.

Camber

- Tilt of the wheel from the vertical when seen from the front. Positive camber - top outward; negative - top inward; zero = vertical.

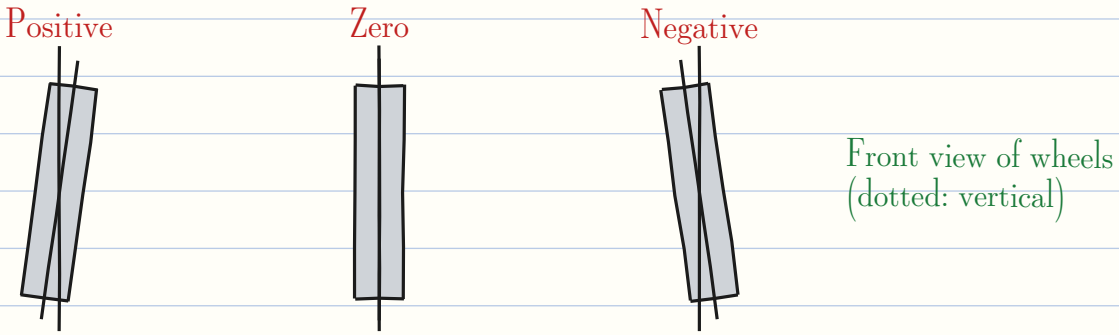


Fig 3.3 Camber angle

- Typical 0.5 deg - 1.5 deg positive. Purpose: (1) bring the tyre contact point under the king-pin -> reduces steering effort; (2) compensate for wear in bearings & flexing of axle under load; (3) prevent wheel falling out/tilt inward.
- Too much +ve camber -> outer tread wear; too much -ve camber -> inner tread wear; unequal camber -> vehicle pulls to side with more +ve camber.

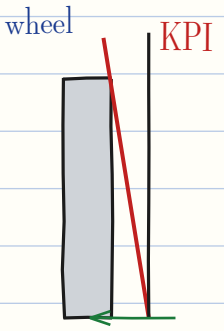
Caster

- Angle between the king-pin axis (steering axis) and the vertical, seen from the side. Positive caster - top of axis tilted backward (towards driver).
- Typical +2 deg to +4 deg (power-steering cars up to +6 deg). Purpose: directional stability + self-centring (return-ability) after a turn - like a caster wheel of a trolley / bicycle fork.
- Too much +ve caster -> heavy steering, shimmy; negative caster -> light steering but wander. Unequal caster -> pulls to side with less +ve caster.

4. KPI, Toe-in, Toe-out & Scrub Radius

King-pin inclination (KPI) / steering axis inclination (SAI)

- Inward tilt of king-pin top, seen from the front; 6 - 8 deg.



KPI: (1) reduces scrub radius -> less steering effort (2) returns wheel to straight ahead (lifts vehicle when wheel turned -> self-centring)

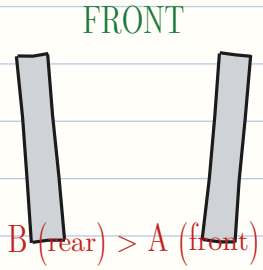
Included angle = Camber + KPI

Fig 3.4 King-pin inclination and scrub radius

- Scrub radius = distance between king-pin axis & tyre centre line at ground. Large +ve scrub -> kick-back on bumps; zero scrub -> less effort; negative scrub (modern FWD) -> stable braking.

Toe-in

- Front wheels are set so that the front edges are closer than the rear edges (distance A < B). Typical toe-in 2 - 4 mm (0 - 3 mm).



Toe-in = B - A

Purpose: offsets the tendency of +ve camber wheels to roll outward; keeps wheels parallel when rolling

Fig 3.5 Toe-in (top view)

- Toe-out on turns (turning angle): the inner wheel turns more than the outer (Ackermann) giving ~ 20 deg inner vs 18 deg outer - also tests bent arms.
- Wrong toe -> feathered tyre edges, rapid tread wear, wander.

5. Wheel Alignment & Balancing

- Wheel alignment = setting camber, caster, toe (and thrust angle) to manufacturer specification so that wheels are parallel / properly angled to each other and to the vehicle frame.

Symptoms of poor alignment

- Vehicle pulls to one side; steering wheel off-centre; uneven / rapid tyre wear; heavy or loose steering; vibration / shimmy; squealing tyres in turns; poor return-ability.

Causes

- Hitting kerb / pothole, worn ball joints, tie-rod ends or bushes, bent arms, sagging springs, changed ride height, accident damage.

Alignment procedure (4-wheel)

1. Check tyre pressure, wear and wheel run-out; inspect suspension & steering joints for play.
2. Place vehicle on level alignment rack; fit sensor heads / clamps (laser / camera / CCD) to all wheels; compensate run-out.
3. Read caster, camber, KPI, toe on the screen versus manufacturer data; adjust using eccentric bolts, shims, tie-rod sleeves.
4. Set toe last, centre the steering wheel, road test.

Wheel balancing

- Uneven mass around tyre-wheel assembly -> vibration. Static balance (single plane: wheel hops) & dynamic balance (two planes: wheel wobbles / shimmy). Corrected using lead / clip-on weights on rim edges, found on a computerised balancer.

Angle	Typical value	Main effect
Camber	+0.5 to +1.5 deg	Tyre wear, pull, steering effort
Caster	+2 to +4 deg	Stability, self-centring
KPI / SAI	6 - 8 deg	Self-centring, scrub radius
Toe-in	2 - 4 mm	Tyre wear, straight running

6. Steering Linkages

- The linkage transmits motion from the steering gearbox to the two steering knuckles so that both wheels turn through the correct angles.

Components

- Pitman arm (drop arm) - swings from the gearbox cross shaft.
- Drag link (steering link) - pitman arm to steering arm (long link).
- Steering arm / knuckle arm - fixed to stub axle (knuckle), rotates it on king-pin.
- Tie rod (track rod) with adjustable sleeve for toe setting; tie-rod ends (ball joint).
- Idler arm - supports the centre link (parallelogram linkage).
- Steering column & universal joint, collapsible (energy-absorbing) for safety.

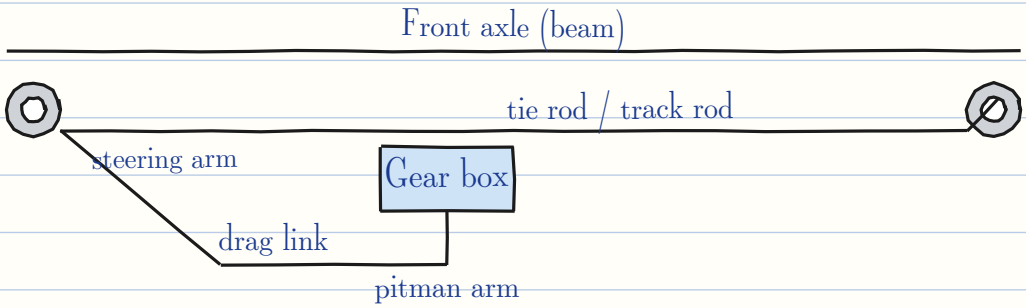


Fig 3.6 Steering linkage for a rigid-axle vehicle

Linkage layout types

- Rigid axle (solid axle): one-piece tie rod + drag link (above).
- Independent suspension: split (three-piece) tie rod - centre link + 2 tie rods + idler arm; or rack & pinion where tie rods are directly joined to the rack ends.
- Ball joints & rubber bushes need lubrication / replacement; worn joints cause free play and wander.

Steering ratio

Steering ratio = steering-wheel rotation / road-wheel swivel angle: 12:1 to 24:1. Higher ratio = lighter effort but more turns lock-to-lock (3-4 turns for manual, 2-3 for power steering).

7. Steering Gearboxes - Types

- The steering gearbox (a) converts rotary motion of the steering wheel into linear / arc motion of linkage; (b) gives mechanical advantage (gear ratio 12:1 - 24:1); (c) must be irreversible or semi-reversible (road shocks not felt at wheel).

Common types

Type	Mechanism	Use
Worm & sector	Worm gear meshes with a sector gear	Older cars / trucks
Worm & roller	Roller on cross-shaft rolls on worm (less friction)	Light trucks
Cam & lever	Cam (worm) with studs / lever bolt	Cars (Ross type)
Recirculating ball (nut)	Balls roll between worm & nut	Trucks, SUVs, buses
Rack & pinion	Pinion drives toothed rack	Most modern cars (FWD)

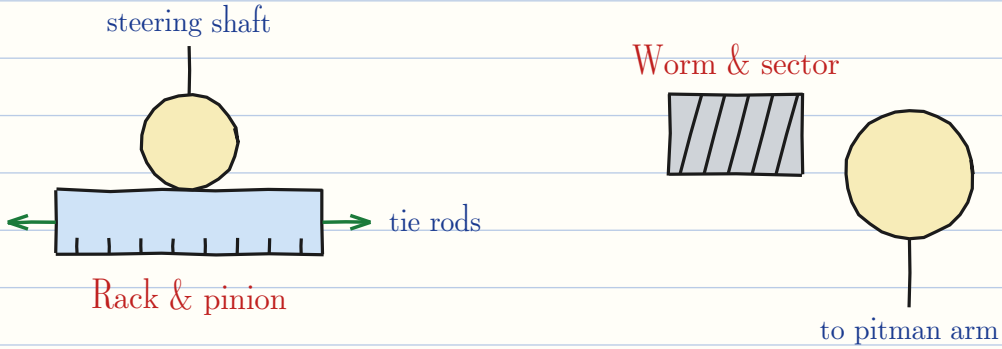


Fig 3.7 Rack & pinion and worm & sector

- Steering ratio = (steering wheel angle) / (pitman arm angle). Lower ratio -> quick response (sports cars); higher ratio -> less effort (heavy vehicles).

8. Recirculating Ball & Rack-Pinion

Recirculating ball steering gear

- A worm shaft with spiral groove carries a ball nut with matching groove; steel balls run in the groove, reducing friction (efficiency ~ 85-90 %). Balls leaving one end are returned through ball return guides (tubes) - recirculate.
- Ball nut has rack teeth meshing with a sector shaft -> sector (cross) shaft rotates -> pitman arm. Adjusters: end-thrust (worm bearings) and sector-lash screw.

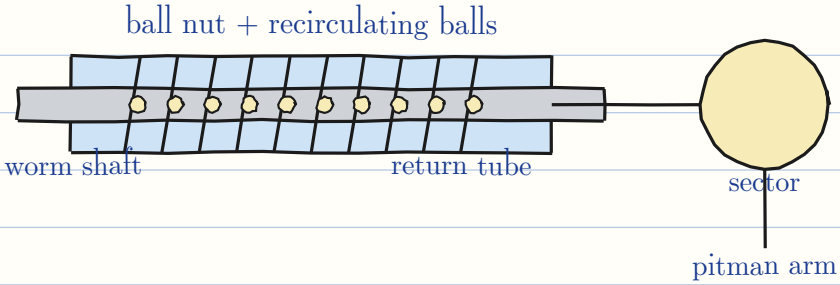


Fig 3.8 Recirculating-ball steering box

Rack & pinion steering gear

- Pinion on the steering shaft meshes with a toothed rack; rotation of the pinion slides the rack left / right; tie rods at the ends move the wheel arms directly -> no pitman arm, drag link, idler.
- Advantages: simple, compact, light, low cost, positive and quick response, good road feel, fewer joints. Disadvantages: transmits road shocks / kick-back, not suitable for heavy axle loads; can be damaged by impact.
- Used in almost all modern small & medium cars (MacPherson suspension); power-assisted by hydraulic or electric motor.

Point	Recirculating ball	Rack & pinion
Parts	More (box, arm, links)	Few
Feel	Isolated, less feedback	Direct, sporty
Load capacity	High	Moderate
Use	Trucks, jeeps	Cars

9. Power Steering

- Power steering supplements the driver effort with hydraulic or electric force -> light steering at parking speeds, still gives feel at speed. If assist fails, the vehicle can still be steered manually (fail-safe).

Types

- Hydraulic power steering (HPS) - engine-driven vane pump (power-assist 70-100 bar), fluid reservoir, rotary control valve, power piston / cylinder, hoses.
- Electro-hydraulic (EHPS) - pump driven by electric motor (saves fuel).
- Electric power steering (EPS) - electric motor on column / rack; torque sensor + ECU (see next page).

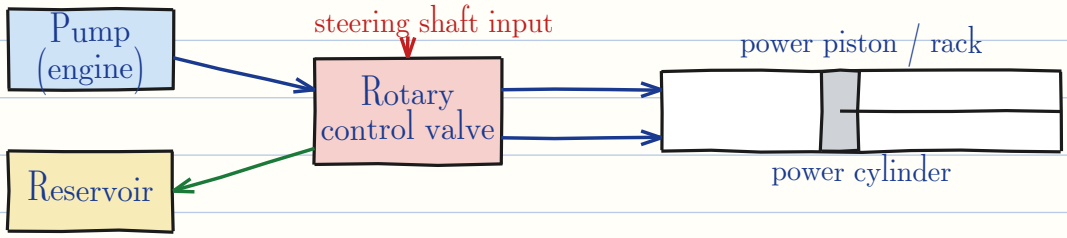


Fig 3.9 Hydraulic power steering (integral rack type)

Working

- When the driver turns the wheel, a torsion bar in the rotary valve twists a little -> valve spool opens oil passage to one side of the power piston -> hydraulic force assists the rack; the other side is open to return. When the wheel stops, the torsion bar un-twists, valve centres, assist stops.
- Types by layout: integral (gear + piston in one housing), linkage (cylinder separate), semi-integral.
- Fluid: ATF / special PSF; faults - belt slip, low fluid (groan), air in system, leaking seals / hoses.

10. Electric Power Steering (EPS)

- EPS uses an electric motor (brushed / brushless) to assist steering, controlled by an ECU using signals of steering torque, steering angle and vehicle speed.

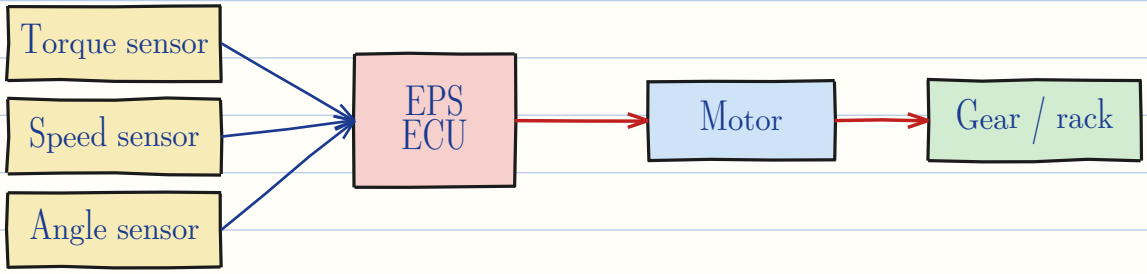


Fig 3.10 Electric power steering (block diagram)

Types by motor location

- Column-assist (C-EPS) small cars; pinion-assist (P-EPS); rack-assist (R-EPS) - bigger vehicles & high loads.

Advantages

- No hydraulic fluid, pump, hoses -> compact, eco-friendly; assist only when needed -> saves 3-5 % fuel; variable assist (heavy at high speed); supports lane-keep, park assist, self-parking; tunable feel.

Disadvantages

- Less natural road feel (improved with software), needs high current (vehicle electrical load), cost for heavy-duty vehicles.

Point	Hydraulic PS	Electric PS
Power source	Engine-driven pump	Electric motor (battery)
Efficiency	Pump runs always (parasitic loss)	On demand only
Maintenance	Fluid, belts, hoses	Almost none
Tuning	Fixed	Software adjustable

Steer-by-wire

Future: no mechanical link - sensors + actuators (steer-by-wire) with redundant ECUs; used in advanced / autonomous prototypes.

11. Front Axle & Stub Axles

- Front axle carries the vehicle weight at the front, resists braking & steering forces, and holds the wheels via stub axles that pivot on king pins.

Types

- Live front axle - also drives wheels (4WD / AWD vehicles). Dead front axle - only supports weight and steering (commonly used).
- Section: I-section / H-section forged steel (medium-carbon, Ni-Cr steel) - drop-forged I-beam is the usual choice; tubular for light vehicles. Ends are shaped for the king-pin boss; centre is dropped to lower the engine.

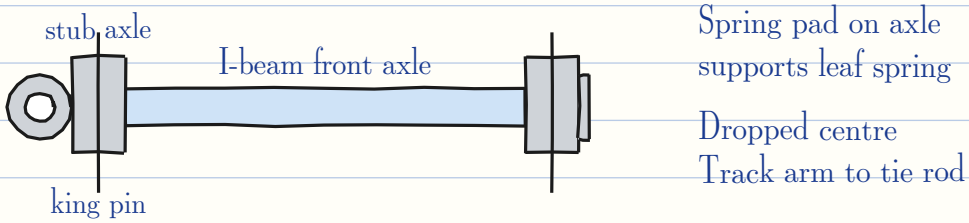


Fig 3.11 Front axle with stub axles

Stub axle arrangements (how the king-pin is connected)

Type	Description
Elliot	Yoke (fork) on the axle end holds stub axle; king-pin fixed in the axle; stub axle has the bearing
Reverse Elliot	Yoke is on the stub axle, axle end fits in between; most common
Lamoine	Stub axle L-shaped, king-pin in axle end, fork on stub axle (opening downwards)
Reverse Lamoine	L-shape stub axle, inverted; king-pin in stub axle

- King-pin bushed with bronze bush; thrust bearing (roller / ball) takes the vertical load; shims set vertical play.

12. Suspension System - Introduction

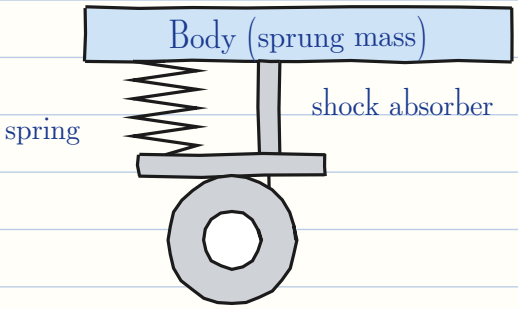
- Suspension = system of springs, shock absorbers and linkages connecting the vehicle body (sprung mass) to the wheels (unsprung mass).

Functions

- Isolate body from road shocks (ride comfort); keep tyres in contact with road (road holding); support weight; maintain correct wheel alignment; resist roll, pitch, dive & squat; transmit driving & braking forces.

Parts

- Springs (leaf, coil, torsion bar, air, rubber), dampers (shock absorbers), control arms / wishbones, links / rods, anti-roll bar, bushes and ball joints.



Sprung mass: body, engine, frame

Unsprung mass: wheels, tyres, axle, brakes -> should be minimum

Sprung / unsprung ratio high = better ride

Fig 3.12 Basic quarter-car suspension

Classification

- Rigid (solid / dependent) axle - both wheels on one axle: leaf spring, coil spring, air; Independent - each wheel moves separately: wishbone, MacPherson, trailing arm, swing axle, multi-link.
- Based on spring: leaf, coil, torsion bar, air, hydro-pneumatic, rubber. Active/semi-active electronic suspension (Unit V).
- Natural frequency for comfort: 1 - 1.5 Hz (60 - 90 cycles/min).

13. Independent Suspension

- Each wheel is connected to the body separately, so a bump on one side does not affect the other wheel.

Advantages

- Better ride & handling; lower unsprung weight; more engine space (no axle); less wheel shimmy; easy to design geometry.

Disadvantages

- Complex, costly, more joints; track & camber change with wheel travel.

Common types

1. Double wishbone (SLA) - upper + lower arms of unequal length; coil spring + damper; good camber control; sports cars, SUVs.
2. MacPherson strut - strut (spring + damper + steering pivot) with a single lower control arm; simple, light, saves space; most FWD cars.
3. Trailing arm / semi-trailing arm - rear suspension; arm pivots at front, carries wheel at rear.
4. Swing axle - half-axles pivoted at differential (VW Beetle); big camber change.
5. Multi-link - 3-5 links per wheel; best compromise comfort/handling; luxury cars.

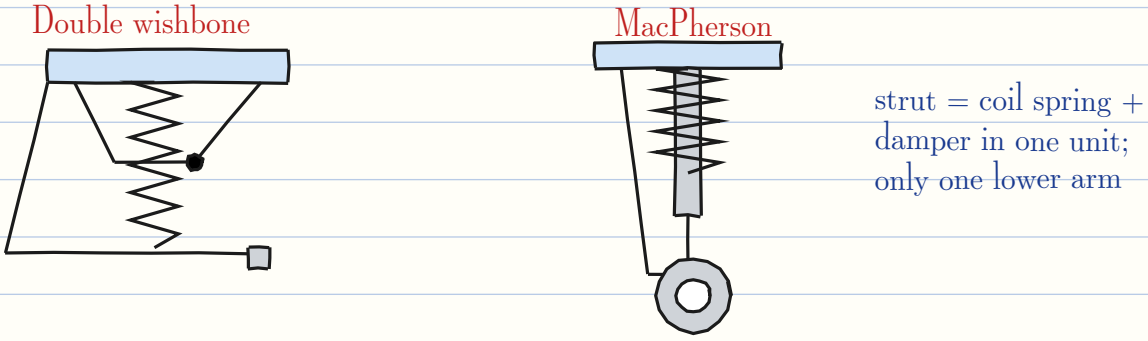


Fig 3.13 Wishbone and MacPherson strut (front view)

14. Solid Axle & Leaf Spring Suspension

- Solid (rigid) axle suspension: both wheels on one beam axle (front or rear), attached to the frame via springs. Used in trucks, buses, jeeps, SUV rear axles.
- Advantages: simple, strong, low cost, camber constant with load, high load capacity. Disadvantages: higher unsprung mass, bump on one wheel tilts the other; poorer ride.

Leaf spring

- A stack of steel plates (leaves) of decreasing length clamped together: semi-elliptic is the commonest form. Longest = master (main) leaf with eyes at both ends (front eye fixed on pin / rear on shackle to allow length change).

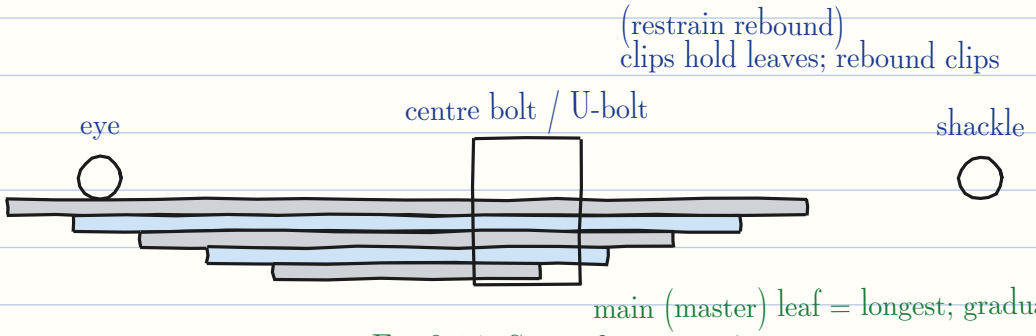


Fig 3.14 Semi-elliptic leaf spring

- Main parts: master leaf, graduated leaves, centre bolt, rebound clips, U-bolts to axle, spring seat, shackle & bushes, inter-leaf friction (damps vibrations, but causes harshness -> use lubricated / plastic liners).
- Material: silico-manganese / chrome-vanadium steel, shot-peened for fatigue life. Types: semi-elliptic, quarter-elliptic, transverse, parabolic (taper) leaf - fewer leaves, lighter, less friction.
- Spring rate k = load / deflection; leaf spring also locates the axle and transmits drive / braking torque (no extra links needed).

15. Coil Spring & Torsion Bar

Coil spring

- Helically wound round steel wire (spring steel, 10-20 mm dia.). Stores energy by torsion of wire. Used in independent front & rear suspension (car, SUV).
- Advantages: lightweight, compact, no inter-leaf friction, durable, wide rate range, progressive-rate (variable pitch / conical) possible. Disadvantage: cannot take side loads -> needs control arms / links.
- Rate: k = G d^4 / (8 D^3 n) (G = shear modulus, d = wire dia, D = mean coil dia, n = active coils).

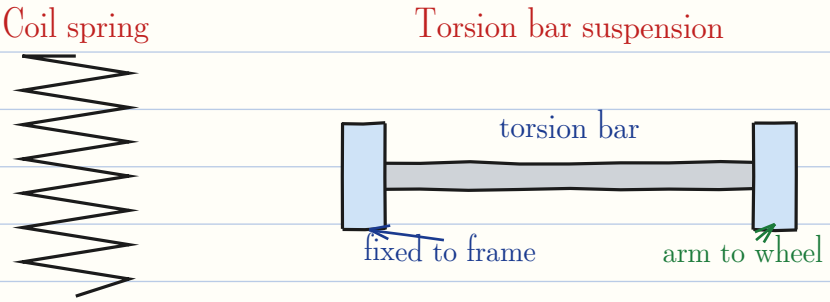


Fig 3.15 Coil spring and torsion bar

Torsion bar spring

- A steel rod fixed at one end to the frame, other end connected to a lower control arm; wheel movement twists the bar and its elasticity gives spring action.
- Pros: compact, adjustable ride height by turning the anchor, light, long life, low cost. Cons: limited deflection (twist), needs well-finished surface; left/right bars not interchangeable (pre-stressed).
- Used in Citroen, VW Beetle, Chrysler (front), many 4WD pickups & Military vehicles.

Rubber spring

- Rubber blocks in compression / shear - give self damping, noise insulation. Used as auxiliary / bump stops, engine mounts, bushes, trailers.

Spring rate & frequency

f = (1/2pi) sqrt(k / m) . Softer spring (low k) gives lower frequency = more comfort but more body roll; stiffer spring = better handling.

16. Air Suspension

- Uses air bags (bellows / air springs) instead of steel springs. Compressed air (4 - 8 bar) carries the load; pressure is varied to control ride height & stiffness.

Main components

- Compressor, dryer + reservoir, height (levelling) valve (linkage to axle), air spring (rolling lobe / bellows / convoluted type), shock absorbers, control ECU with ride-height sensors, solenoid valves.



Load increases -> body sinks -> valve admits air -> body rises (self levelling)

Fig 3.16 Air suspension layout

Advantages

- Constant ride height & frequency irrespective of load; excellent comfort; adjustable (highway / off-road / loading mode); long life of cargo; reduces road damage; used in buses, trucks, trailers, Volvo, Mercedes, Range Rover, Audi.

Disadvantages

- Costly, complex, air leaks, needs compressor & maintenance; limited lateral stiffness (needs linkages).

Types of air springs

- Bellows (convoluted) - 1 to 3 folds, high load capacity, used on trucks / buses. Reversible sleeve - flexible sleeve rolls over a piston. Rolling lobe - compact, long stroke, used in cars (smooth ride).
- Air-over-oil / hydropneumatic (Citroen) - gas spring with oil damper & pump; self-levelling with very soft ride.

Working idea

Spring rate rises automatically with load because the air pressure rises - so the natural frequency stays almost constant from empty to full load.

17. Shock Absorbers & Stabiliser Bar

- Shock absorber (damper) dissipates the spring's stored energy as heat to damp oscillations; without it the body bounces repeatedly. Controls spring rebound & wheel hop.
- Types: friction (old), hydraulic telescopic (double acting) - standard, gas-filled (monotube / twin-tube), adjustable, magnetorheological (Unit V).

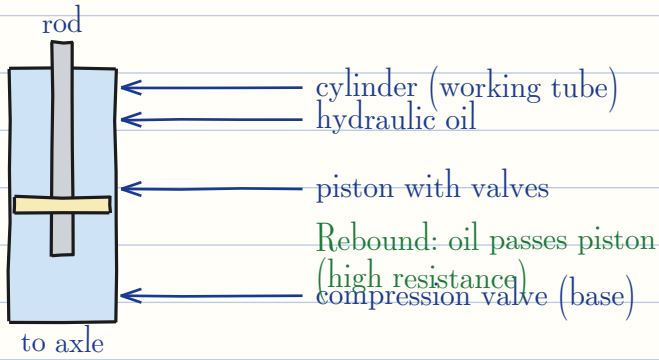


Fig 3.17 Telescopic hydraulic damper (twin tube)

- Working: on compression (bump) oil is forced through small orifices / the base valve; on rebound the piston valve restricts oil flow more (rebound damping ~ 2-3 x compression damping). Energy -> heat -> dissipated through the tube.
- Gas-filled (nitrogen 5-25 bar): prevents oil foaming (aeration) and fade; better response.

Stabiliser (anti-roll) bar

- U-shaped torsion bar across axle linking left & right lower arms; resists body roll in corners by twisting (equalises load on both wheels) without affecting ride in straight line.

Spring	Typical use
Leaf	Trucks, buses, rear of vans / jeeps
Coil	Cars - independent
Torsion bar	Compact 4WD, some cars
Air	Buses, luxury cars, trailers

UNIT IV: Wheels, Tyres & Braking System

Syllabus: Wheels and tyres - construction, type & specification, tyre wear and causes; brakes - need, classification, drum & disc, mechanical, hydraulic, pneumatic, vacuum assist, retarders, anti-lock braking system (ABS).

1. Wheels

- The wheel supports the vehicle weight, transmits driving & braking torque to the road through the tyre, and takes steering & cornering forces. It consists of rim, disc (spider), hub.

Requirements

- Strong yet light (low unsprung mass), true running (low radial & lateral run-out), well balanced, good heat dissipation for brakes, corrosion resistant, easy to remove / fit, cheap.

Types of wheels

1. Disc (pressed steel) wheel - rim welded to a stamped steel disc with holes for ventilation; cheap & strong; cars, light trucks.
2. Wire-spoke wheel - steel spokes (laced) between hub & rim; light, strong, good cooling; motorcycles, vintage / sports cars.
3. Cast alloy wheel - Al / Mg alloy cast or forged; light (up to 25 % less), better heat transfer, style; costlier, can crack on impact. Mg alloy -> racing.

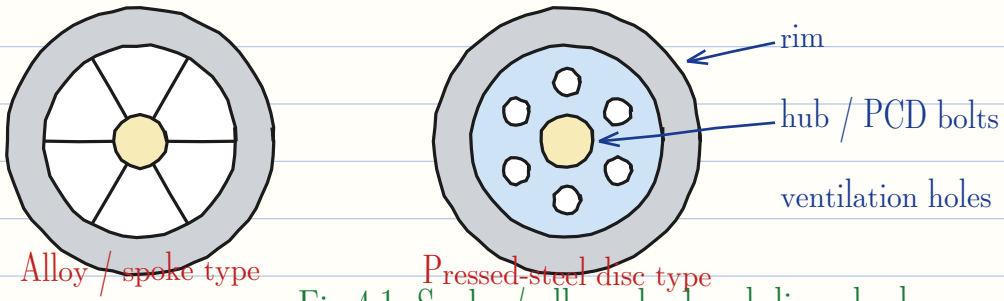


Fig 4.1 Spoke / alloy wheel and disc wheel

Wheel size

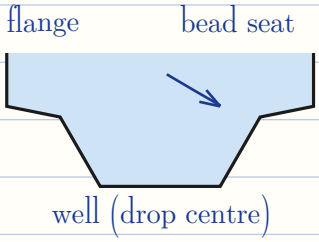
Specified by rim diameter x width, e.g. 15 x 6J (15 inch dia., 6 inch width, J = flange shape), plus PCD (pitch-circle dia. of bolts, e.g. 4 x 100), offset (ET) and centre bore.

2. Rim Construction & Types

- The rim holds the tyre bead; its section is shaped to seat the tyre firmly and (for tubeless) hold air.

Parts of rim

- Flange (side wall support), bead seat (tapered 5 deg - grips bead), well / drop centre (lets the bead drop in during fitting), valve hole, safety hump (tubeless - keeps bead from slipping into well).



Drop-centre rim: most common (cars)
- one piece, well in the middle
- tubeless tyres fit easily

Wide-base / flat-base rim: trucks
- detachable flange, lock ring

Fig 4.2 Drop-centre rim section

Rim types

Type	Feature	Use
Drop-centre (well base)	Deep well in centre for tyre fitting; one piece	Cars, LCVs
Flat-base	Flat seat; demountable flange / lock ring	Trucks, buses
Semi drop-centre	Shallow well	Light trucks
Divided (two-piece)	Halves bolted together	Alloy cars, racing
Multi-piece / split rim	Lock ring, side ring, base	Heavy trucks

Materials

- Steel (pressed from sheet, rolled & welded), Al alloy (cast / forged), Mg alloy. Alloy rims need balancing weights of adhesive type.
- Wheel offset (ET): distance between hub-mounting face & rim centre-line; +ve offset (FWD cars) puts the wheel closer to the arch; affects scrub radius & track.

3. Tyre - Functions & Construction

- A tyre is a flexible toroidal rubber-fabric-steel structure filled with compressed air, mounted on the rim.

Functions

- Carry load (via air pressure); provide traction & braking grip; absorb road shocks (cushion); transmit cornering force; resist wear & heat.

Parts of a tyre

- Tread - thick outer rubber with grooves (pattern) that touches road. Shoulder. Sidewall - protects carcass, carries markings.
- Carcass (casing / plies) - layers of rubberised cord fabric (rayon, nylon, polyester, steel) -> strength & flexibility.
- Belt / breaker - steel / aramid belts below tread (radial) - stiffens tread, resists punctures.
- Bead - bundle of high-tensile steel wire coated with rubber; locks tyre on the rim. Inner liner - air-tight rubber (tubeless). Chafer - protects bead from rim wear.

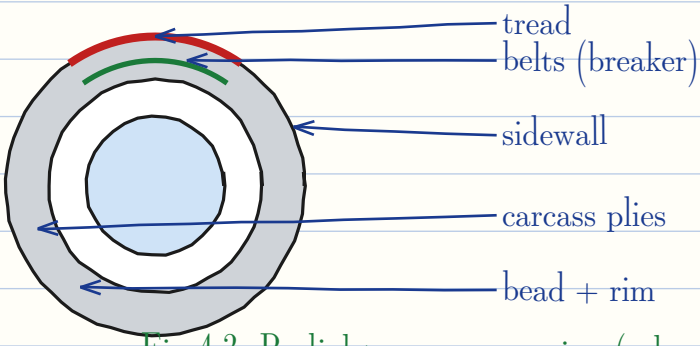


Fig 4.3 Radial tyre cross-section (schematic)

- Tread patterns: rib (highway, low noise), lug (traction / off-road), rib-lug, block / directional, asymmetric; grooves + sipes channel water (prevent aquaplaning).
- Tread compound: natural + synthetic rubber, carbon black, silica, sulphur (vulcanisation), anti-oxidants.

4. Types of Tyres

Based on construction (carcass layout)

1. Cross-ply (bias) tyre - plies at 30-40 deg to the centreline, alternate plies crossed. Stiff sidewall, harsh ride, heat build-up, shorter life; cheap; trucks (older), agri.
2. Belted bias tyre - bias plies + circumferential belts under tread -> better tread life & stability than bias.
3. Radial-ply tyre - plies at 90 deg (radial) to the centreline + steel belts under tread. Flexible sidewall, large contact patch, low rolling resistance, cooler, longer life (30-40 % more), better grip & fuel saving; standard for cars today.

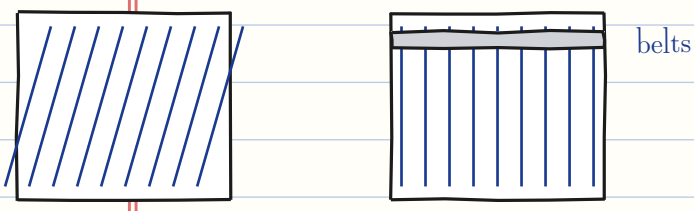


Fig 4.4 Cord direction in cross-ply and radial tyres

Based on air containment

- Tube-type - separate inner tube with valve; puncture -> sudden deflation. Tubeless - inner liner + airtight bead seat, valve on rim; punctures leak slowly, lighter, cooler, safer, easy repair (plug).

Other types

- Run-flat (reinforced sidewall, run 80 km at 80 km/h flat), snow / winter, all-season, off-road (M+S, A/T, M/T), low-profile sports (UHP), space-saver spare.

Point	Cross-ply	Radial
Ride	Harsh	Smooth
Tread life	Shorter	Longer (+30-40 %)
Rolling resistance	High	Low (fuel saving ~3-5 %)
Cost	Low	Higher

5. Tyre Specification & Designation

- Tyre sidewall carries a code describing size, type, load & speed rating. Example: 205 / 55 R 16 91 V.

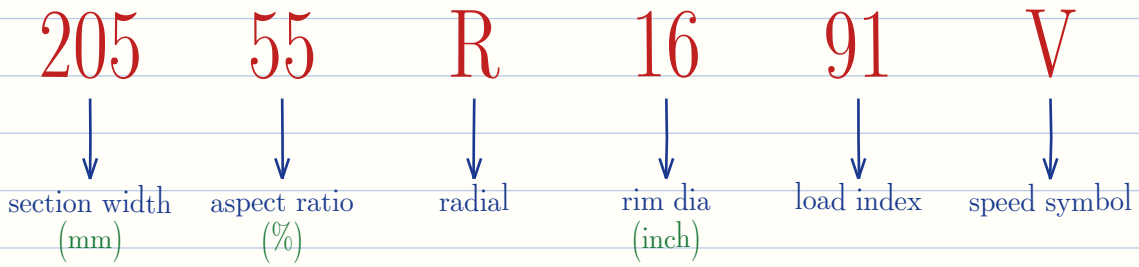


Fig 4.5 Reading a tyre marking

Terms

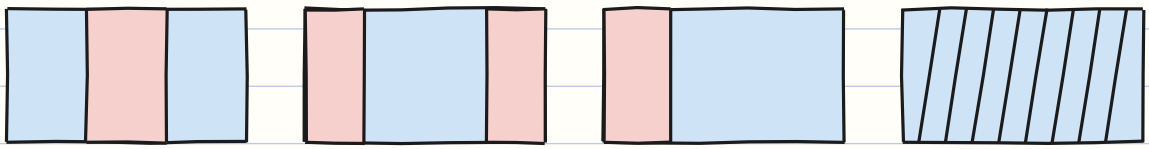
- Section width: widest sidewall-to-sidewall in mm. Aspect ratio (profile): sidewall height / section width x 100 (55 -> sidewall height = 0.55 x 205 = 113 mm); lower = sportier, stiffer.
- R = radial; D = diagonal / bias; ZR = high-speed radial. Rim dia in inches. Old bias format: 6.00 - 16 (width in inch - rim dia).
- Load index (LI): 91 = 615 kg per tyre. Speed symbol: L = 120, N = 140, P = 150, Q = 160, S = 180, T = 190, H = 210, V = 240, W = 270, Y = 300 km/h.

Marking	Meaning
Tubeless / TL	Tubeless tyre
M+S	Mud & snow capability
DOT code	Manufacture plant + week + year (e.g. 2524 = week 25 of 2024)
Treadwear / Traction / Temp	UTQG grades (US): e.g. 400 A A
Max. pressure / load	Printed on sidewall (kPa / psi, kg)

- Recommended inflation pressure: 28 - 36 psi (190 - 250 kPa) for cars; check cold; TPMS warns of low pressure.

6. Tyre Wear & Causes

Tyre wear patterns act as a diagnosis tool - they show what is wrong with inflation, alignment or suspension.



Centre wear — over-inflation | Both-edges wear — under-inflation | One-edge wear — camber | Feathering — toe error

Fig 4.6 Tread-wear patterns (red = worn area)

Wear pattern	Cause	Remedy
Centre worn	Over-inflation	Correct pressure
Both shoulders worn	Under-inflation, overloading	Inflate; reduce load
One shoulder	Excess camber / worn arms	Align; fix suspension
Feathered edge	Wrong toe-in / toe-out	Set toe
Cupping / scalloping	Worn shocks, imbalance	Replace dampers; balance
Flat spots	Locked wheel braking	Use ABS / smooth braking
Diagonal wear	Misalignment, worn bushes	Rotate, align

General causes & prevention

- Hard braking & acceleration, fast cornering, high temperature, rough roads, mixing different tyres, long storage / UV ageing (cracking).
- Tyre rotation every 8,000 - 10,000 km (cross pattern for radials); check pressure monthly; wear indicators (TWI bars at 1.6 mm); replace tread < 1.6 mm; retreading (hot / cold) for trucks.

7. Brakes - Need & Classification

- Brake = device that slows down or stops a vehicle (and holds it stationary) by converting its kinetic energy into heat through friction.

Need / functions

- Reduce speed on demand, stop in shortest distance safely & in a straight line, hold vehicle on slopes (parking), permit higher average speeds.

Requirements

- Effective, progressive (pedal effort proportional to brake force), balanced (front / rear / left / right), minimum fade, quick release, small pedal effort, durable, silent, fail-safe (dual circuit).

Classification

Basis	Types
Purpose	Service (foot) brake, parking (hand) brake, emergency
Method of actuation	Mechanical (rods / cables), hydraulic, pneumatic (air), vacuum-assisted, electric (e-brakes), electromagnetic
Brake unit	Drum (shoe) brake, disc brake, band brake (old transmission brakes)
Location	Wheel brakes, transmission brake (on propeller shaft)
Operation	Foot-operated, hand-operated, automatic (trailer breakaway)
Extra systems	Retarders (exhaust, hydraulic, electric), ABS, EBD

Braking dynamics

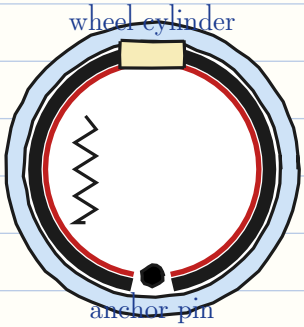
- Retardation a = F / m ; stopping distance s = v^2 / (2 mu g) (v in m/s, mu = tyre-road friction 0.7-0.9 dry). Total = reaction distance + braking distance.
- Weight transfer to the front during braking -> front brakes do ~ 70 % of work -> bigger front discs; rear brake force limited by a proportioning valve to avoid rear lock-up.

Brake fade

Loss of braking effectiveness caused by overheating of linings / fluid (vapour). Disc brakes fade less than drum brakes.

8. Drum Brakes - Construction

- A rotating drum (bolted to the wheel hub) is pressed from the inside by two curved brake shoes with friction linings, mounted on a stationary back plate (backing plate).



Brake drum (cast iron, rotates)
Brake shoes + linings
Return (retracting) springs
Wheel cylinder (hydraulic) / cam
Back plate + anchor pin
Self-adjuster / hand-brake lever

Fig 4.7 Drum brake (leading-trailing type)

Working

- Pedal pressure -> wheel cylinder pistons push shoes outward against the rotating drum -> friction slows the wheel. Release -> return springs pull shoes back.

Materials

- Drum: grey cast iron (sometimes with steel liner / Al fins); Shoes: pressed steel / Al; Lining: organic (non-asbestos), semi-metallic, ceramic - riveted / bonded; friction coefficient 0.35 - 0.45.

Self-energising (servo) action

- When the drum rotation drags the shoe in the direction of rotation ("leading shoe "), friction wedges it further against the drum -> extra braking force without more pedal effort. Trailing shoe is pushed away -> no self-energisation.
- Disadvantages: heat retained inside drum -> fade, water retention, adjustment needed; advantages: cheap, strong parking brake, protected from dirt.

9. Drum Brake Types & Mechanical Brakes

Type	Arrangement	Remark
Leading-trailing	One wheel cylinder; one leading + one trailing shoe (same anchor)	Equal in both directions; rear brakes of cars
Two leading shoe	Two single-piston cylinders; both shoes are leading in forward motion	Strong forward braking, poor in reverse; front brakes (older)
Duo-servo	Both shoes linked (floating); primary shoe pushes secondary	Highest self-energisation; US cars, trucks
Single anchor / Twin leading	Two anchors (two pivots)	Motorcycles, scooters (mechanical)

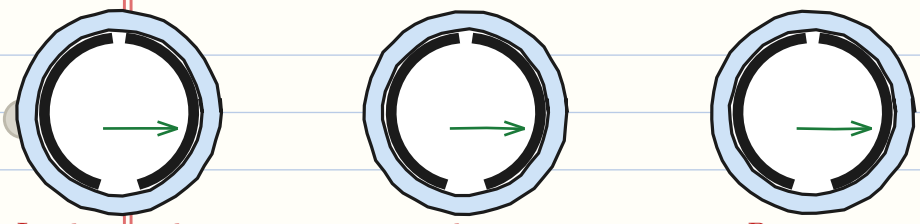


Fig 4.8 Drum brake arrangements

Mechanical brake linkages

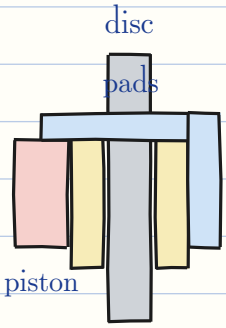
- Pedal / hand lever -> rods / cables / levers / cam or wedge -> shoes. Equaliser makes the force same at each wheel.
- Used for parking (hand) brake even in hydraulic cars: cable -> lever in rear shoe -> expands shoes (duo-servo / drum-in-disc). Also motorcycle rear brake, old cars.
- Disadvantages: friction & lost motion in linkage, unequal braking, frequent adjustment, hard to apply large force.
- Cam-operated (S-cam) brakes in air brakes: S-cam spreads shoes, turned by slack adjuster.

Brake adjustment

Self-adjusters (ratchet / star-wheel) maintain shoe-to-drum clearance 0.2 - 0.4 mm; manual adjust where no auto adjuster; slack adjuster in air brakes.

10. Disc Brakes

- A flat disc (rotor) is bolted to the wheel hub; a caliper straddles it holding two friction pads that squeeze (clamp) the disc from both sides when hydraulic pressure acts on the caliper piston(s).



Disc / rotor (cast iron, vented)
Brake pads (friction material)
Caliper (piston + seals)
Hydraulic inlet / bleed screw
Dust boot, retaining pins

Fig 4.9 Disc brake assembly (schematic)

Types of caliper

- Fixed caliper: rigid, piston(s) on both sides (2 - 8 pots); better stiffness, more costly - sports/heavy cars.
- Floating (sliding) caliper: pistons on inboard side only; caliper slides on guide pins to apply the outer pad; cheap, compact - most cars.

Disc types

- Solid, ventilated (internal fins), drilled / slotted (gas & water clearing), carbon-ceramic (racing). Pad: semi-metallic, ceramic; wear sensor / squealer.

Point	Disc	Drum
Heat dissipation	Excellent (open)	Poor (enclosed)
Fade / wet performance	Better	Worse
Self-energisation	None (needs servo)	Yes
Weight / adjust	Light, self-adjusting	Heavier, manual adj.
Use	Front (all), rear on cars	Rear (economy), trucks

11. Hydraulic Brake System

- Based on Pascal's law - pressure applied to a confined fluid is transmitted equally in all directions. Pedal force -> master cylinder -> fluid pressure (up to 100-150 bar) -> wheel cylinders / calipers.

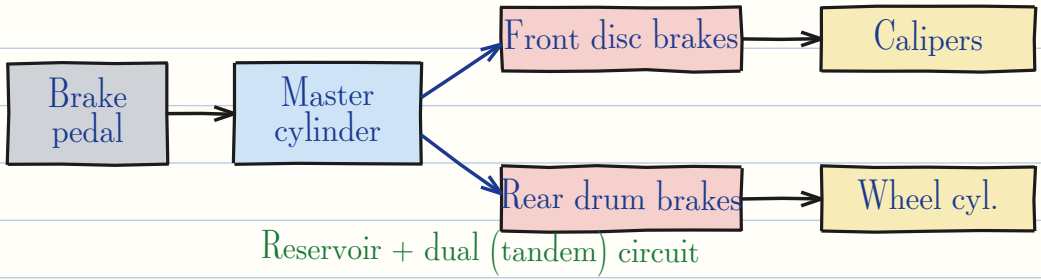


Fig 4.10 Hydraulic brake layout (diagonal / front-rear split)

Components

- Brake pedal (lever ratio 4:1 - 6:1), master cylinder (tandem) + reservoir, brake lines (steel pipes, flexible hoses), wheel cylinders (drum) / caliper pistons (disc), proportioning / pressure-limiting valve, brake switch.

Working

1. Pressing pedal pushes master piston -> fluid pressure rises in the line.
2. Same pressure reaches all wheel pistons; force at wheel = pressure x piston area -> shoes / pads apply.
3. On release, return springs push pistons back; fluid returns to master cylinder (residual pressure valve ~ 0.5 bar keeps cups sealed).

- Advantages: equal pressure to all wheels, fast action, less friction, easy layout, self-lubricating, high force multiplication. Disadvantages: air in system (spongy pedal), leak = complete loss (so dual circuit), fluid absorbs moisture.

12. Master Cylinder, Fluid & Bleeding

Tandem (dual-circuit) master cylinder

- Two pistons in one bore - primary (pushrod-operated) and secondary (floating) - each feeds its own circuit (front / rear or diagonal). If one circuit leaks, the other still brakes (legal requirement).

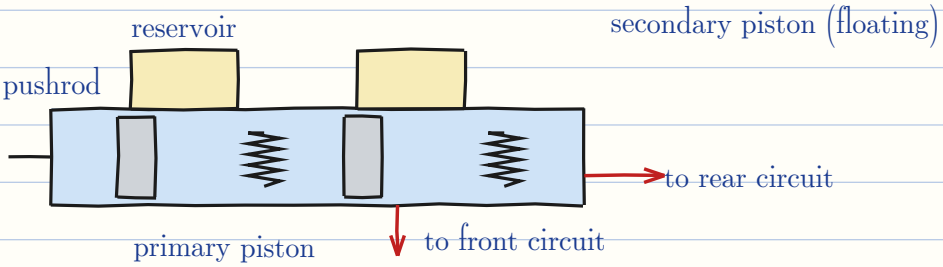


Fig 4.11 Tandem master cylinder

Brake fluid

- Glycol-ether based (DOT 3, DOT 4, DOT 5.1); DOT 5 = silicone (do not mix). Properties: high boiling point (DOT 3: 205 deg C dry; DOT 4: 230 deg C), low freezing point, no corrosion, compatible with rubber seals, good lubricity.
- Hygroscopic - absorbs moisture -> lowers boiling point -> vapour lock / fade; change fluid every 2 years.

Bleeding of brakes

- Removes air (compressible) from lines: open bleed screw while pedal is pressed (manual / pressure / vacuum bleeding) in order farthest wheel from master cylinder first; keep reservoir topped up.

Fault	Likely cause
Spongy pedal	Air in lines; low fluid
Pedal sinks slowly	Master cylinder cups leaking
Pulling to one side	Stuck caliper / contaminated lining
Dragging brakes	Seized piston, faulty return spring
Squeal / judder	Glazed pads, warped disc

13. Pneumatic (Air) Brakes

- Use compressed air (6 - 8 bar) to apply brakes - standard on heavy trucks, buses, trailers (high force, light pedal effort, long vehicles / trailers coupling).

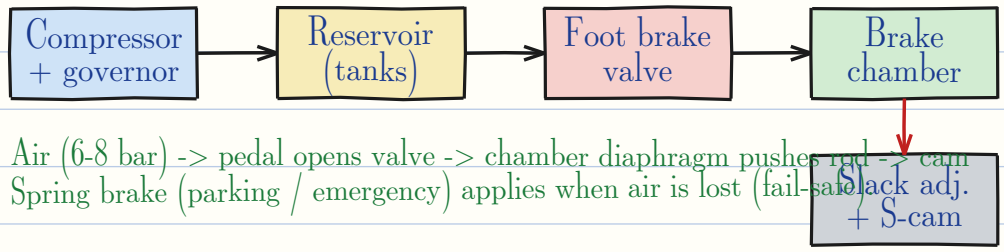


Fig 4.12 Air brake layout (simplified)

Main components

- Compressor (engine-driven), governor (cut-in 6.5 / cut-out 8.5 bar), air dryer, reservoirs (wet / primary / secondary), foot brake valve (treadle), relay valve (quick action at rear), brake chamber (diaphragm), slack adjuster, S-cam, quick release valve, pressure protection valve, trailer gladhands (couplings).

Working

- Pedal -> treadle valve admits air from tank to brake chambers -> push rod -> slack adjuster -> S-cam -> shoes press drum. Release -> air exhausts through quick release valve; return springs retract shoes.
- Spring brakes (park / emergency): powerful spring applies the brake when air pressure falls below ~ 4 bar.

Point	Advantage	Disadvantage
Air brake	Unlimited air source, light pedal, fail-safe, trailers easy	Delay in response (lag), bulky, costly
Hydraulic	Fast, compact	Limited force, leaks

14. Vacuum Assisted Brakes (Servo)

- A vacuum servo (booster) multiplies the driver's pedal force 2-4 times using engine intake manifold vacuum (diesel: vacuum pump) acting on a large diaphragm. Used with hydraulic brakes in cars & LCVs.

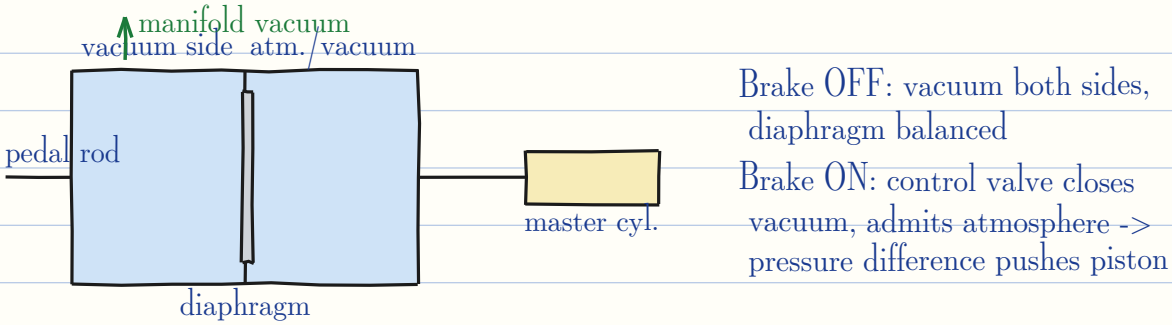


Fig 4.13 Vacuum servo (brake booster)

Working

1. Released: both sides of diaphragm connected to manifold vacuum -> no net force.
2. Pedal pressed: pedal rod moves the valve plunger -> vacuum port closed, atmospheric air admitted to the rear chamber.
3. Pressure difference (atm - vacuum = ~0.7 bar) x diaphragm area gives extra thrust to the master cylinder pushrod.
4. Fail-safe: if vacuum fails, the pedal rod still acts mechanically on the master cylinder (needs higher effort); a check valve retains vacuum for 2-3 assisted brakes.

Types

- Direct-acting booster (on firewall, most cars), tandem booster (two diaphragms, trucks), hydro-vac (hydraulic booster), electric vacuum pump for diesel / EV, brake-by-wire (iBooster).

15. Retarders

- A retarder is an auxiliary braking device that absorbs energy continuously without friction brakes - controls speed on long descents, prevents brake fade, saves brake wear (40-50 %). Common on heavy trucks, buses, hill-area vehicles.

Types of retarders

1. Exhaust brake: butterfly valve in the exhaust pipe closes -> back pressure on the exhaust stroke -> engine acts as a compressor and slows down. Simple, cheap (~ 30-50 % of engine power).
2. Engine (compression release / Jake) brake: exhaust valves are opened near TDC at the end of compression -> compressed air released (not returned to the piston) -> strong braking (up to engine rated power); noisy.
3. Hydraulic retarder (hydrodynamic): a rotor on the driveshaft churns oil against a fixed stator inside a housing; oil friction produces braking torque, heat removed by the engine cooling circuit; torque proportional to speed^2 (Voith, ZF).
4. Electric / eddy-current retarder: steel rotor discs rotate between electromagnet coils on the frame; eddy currents in discs resist rotation -> braking, heat dissipated by air-cooled discs (Telma). No contact, no wear; stages by coil current.

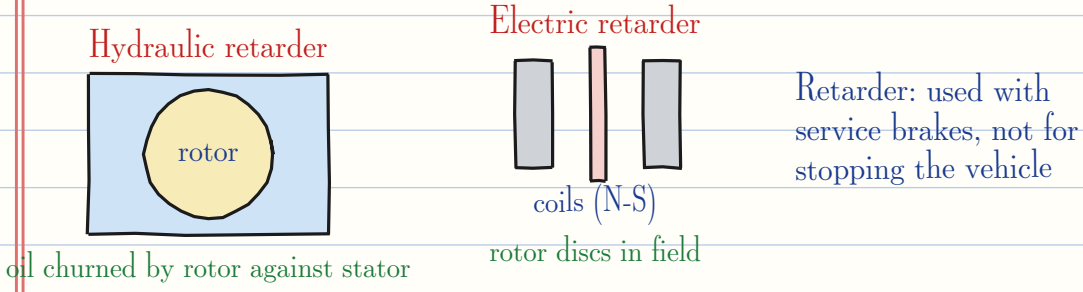


Fig 4.14 Hydraulic and electric retarders

- Retarders reduce speed but cannot hold a vehicle at rest.

16. Anti-lock Braking System (ABS)

- ABS prevents the wheels from locking during hard braking by automatically modulating brake pressure many times a second -> vehicle keeps steering control and shortest possible stopping distance on most surfaces.

Need

- A locked wheel skids (sliding friction < peak friction), loses cornering grip (cannot steer), wears flat spots on tyres. Maximum braking happens at 10 - 20 % slip.

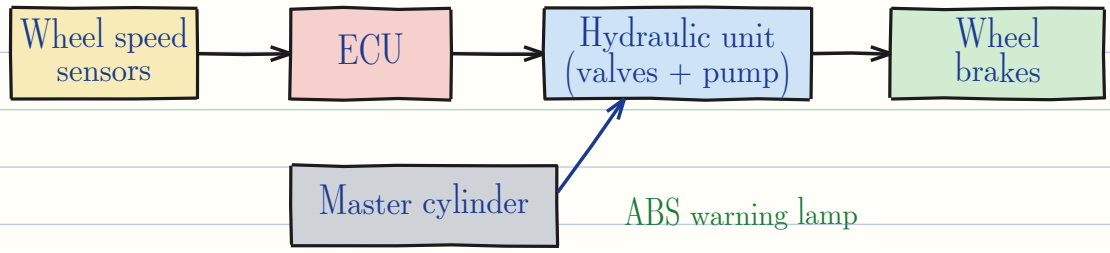


Fig 4.15 ABS block diagram

Components

- Wheel speed sensors (inductive / Hall) + toothed tone ring; ECU; hydraulic modulator - inlet (normally open) & outlet (normally closed) solenoid valves, accumulator & return pump; warning lamp.

Working cycle (3 phases)

1. Pressure build / hold: normal braking; if sensor shows a wheel decelerating too fast (about to lock), ECU closes inlet valve - pressure is held.
2. Pressure release: ECU opens outlet valve; fluid goes to accumulator, pump returns to master cylinder -> wheel speeds up.
3. Pressure re-apply: inlet valve reopens; the cycle repeats up to 15-20 times / s (pedal pulsates).

- Channels: 4-channel 4-sensor (best), 3-channel, 1-channel (rear, pickups). Two-wheeler ABS: 1 / 2 channel.

17. ABS Advantages & Brake Troubleshooting

Advantages of ABS

- Steering control kept under panic stops; shorter stopping distances on wet / slippery roads (on loose gravel / snow may be longer); no tyre flat-spotting; stable vehicle; basis for EBD, TCS, ESP, Brake Assist.

Limitations

- Costly; does not defy physics (speed, bad tyres); pedal pulsation startles drivers (do not pump - press hard and hold); may lengthen stopping distance on soft surfaces.

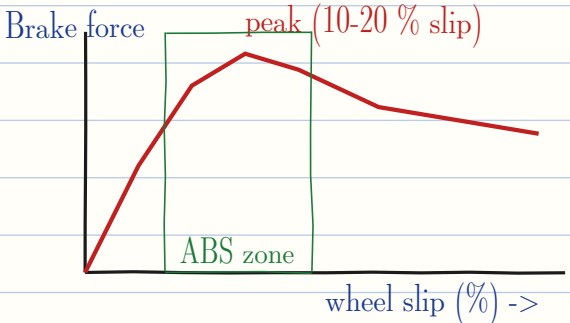


Fig 4.16 Friction vs wheel slip

Brake system troubleshooting

Symptom	Cause	Remedy
Weak brakes	Worn linings, low fluid, air, faulty servo	Replace, bleed, check booster
Brake pull	Uneven pads, tyre pressure, stuck caliper	Service caliper / equalise
Squealing	Glazed / worn pads, no shim	Replace pads, grease backs
Vibration	Warped disc / drum out-of-round	Resurface or replace

Servicing hint

Replace pads in axle pairs; check disc thickness (min mark), bleed in correct order, never reuse brake fluid, and test-drive after service.

UNIT V: Electrical Systems & Advances

Syllabus: Battery, general electrical circuits, Active Suspension System (ASS), Electronic Brake Distribution (EBD), Electronic Stability Program (ESP), Traction Control System (TCS), GPS, hybrid vehicle, fuel cell.

1. Automobile Battery

- The battery is an electrochemical device that stores electrical energy as chemical energy and supplies it to the starter motor, ignition, lights, ECU & accessories when the engine is off or when demand exceeds alternator output.

Functions

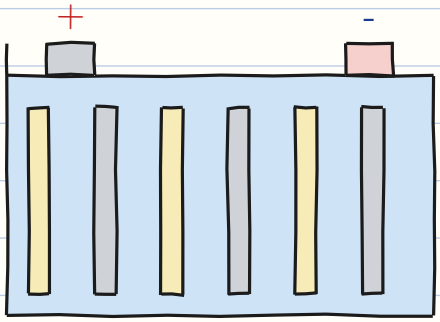
- Crank the engine (starter 100-300 A); supply power with engine off; stabilise voltage (acts as a capacitor for the electrical system); supply extra current at idle.

Types (automotive)

- Lead-acid (SLI: Starting-Lighting-Ignition) - flooded, VRLA (AGM / gel), maintenance-free; Li-ion / LFP (EV, start-stop, bikes); Ni-MH (hybrids); Ni-Cd (old).

Lead-acid battery construction

- Container (polypropylene) with 6 cells x 2.1 V = 12.6 V; each cell = positive plates (PbO_2, brown) + negative plates (spongy Pb, grey) interleaved with separators (microporous PE / glass mat); electrolyte = dilute sulphuric acid (H_2SO_4, sp. gr. 1.28).



+ve plate: lead dioxide
-ve plate: spongy lead
separators (insulator)
electrolyte: dil. H_2SO_4
cell connectors, vent plugs

Fig 5.1 Lead-acid battery (section)

Rating

Capacity in Ah (e.g. 45 Ah @ 20-h rate); CCA (cold cranking amps at -18 deg C); reserve capacity (RC) in minutes.

2. Battery - Chemistry & Maintenance

Chemical reactions

- Discharge: $PbO_2 + Pb + 2H_2SO_4 \rightarrow 2PbSO_4 + 2H_2O$ (both plates turn to lead sulphate; acid becomes weaker, specific gravity falls).
- Charge: $2PbSO_4 + 2H_2O \rightarrow PbO_2 + Pb + 2H_2SO_4$ (reverse; acid strength rises; gassing near full charge: $H_2 + O_2$).

SG vs state of charge

1.280	1.220	1.150
Fully charged	~ 50 % charged	Discharged

measured with hydrometer; temperature correction 0.007 per 10 deg C

Fig 5.2 Specific gravity indicates charge

Charging

- Alternator charges at 13.8 - 14.4 V (regulated). External charging: constant-current (C/10 = 4.5 A for 45 Ah) or constant-voltage; fast charging only for short time. Avoid sparks - hydrogen gas is explosive.

Maintenance

- Keep electrolyte level above plates with distilled water (never acid); clean & grease terminals (sulphation); secure battery; check SG monthly; don't leave discharged (sulphation).
- Sulphation: hard $PbSO_4$ crystals on plates (long discharge) -> low capacity. Self-discharge 3-4 % per month. Life 3-5 years.
- Maintenance-free batteries use Ca-alloy grids (low gassing), sealed with magic-eye hydrometer. Connect +ve first, -ve last (remove opposite). Series connection increases voltage, parallel increases capacity.

3. Other Batteries (Li-ion, Ni-MH, VRLA)

Battery	Cell voltage	Energy (Wh/kg)	Features / use
Lead-acid flooded	2.1 V	30 - 40	Cheap, high surge, heavy; SLI
VRLA (AGM / gel)	2.1 V	30 - 45	Spill-proof, start-stop vehicles, UPS
Ni-Cd	1.2 V	40 - 60	Durable, toxic Cd; old tools
Ni-MH	1.2 V	60 - 100	Hybrid cars (Prius), safe, memory effect low
Li-ion (NMC / NCA)	3.6 - 3.7 V	150 - 250	EV, high energy, need BMS
LFP (LiFePO4)	3.2 V	90 - 160	Long life (3000 cycles), safe; buses, 2W

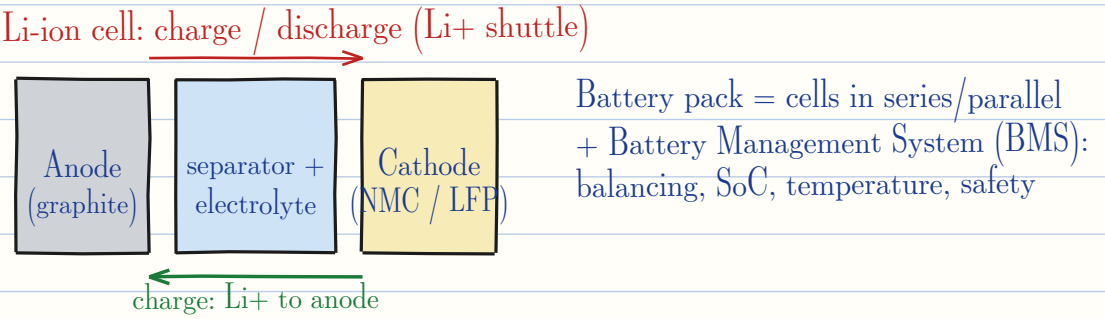


Fig 5.3 Lithium-ion cell principle

Comparison of lead-acid vs Li-ion

- Li-ion: 3-5x energy density, longer cycle life, faster charging, low self-discharge, but costly, thermal runaway risk, needs BMS.
- Lead-acid: low cost, recyclable (> 95 %), robust, high cold-cranking current; heavy, short life.
- Ultracapacitors (EDLC) - high power, quick charge / discharge; used with batteries for regenerative braking & start-stop.

48 V mild hybrid

Uses a 48 V Li-ion battery along with 12 V lead-acid for e-boost, regeneration & stop-start - cheaper than full hybrid.

4. General Electrical Circuits

- Vehicle electrical system is a 12 V (24 V for trucks) DC, negative earth (chassis return) system. Main groups: charging, starting, ignition, lighting, instrumentation, accessories, body electronics (BCM), ECU network (CAN).

Circuit elements

- Source (battery / alternator), conductor (copper wire / wiring harness, colour coded, gauge by current), switch, load (lamp, motor), protection (fuse, fusible link, circuit breaker), relay (low-current switch controlling heavy load), earth return.

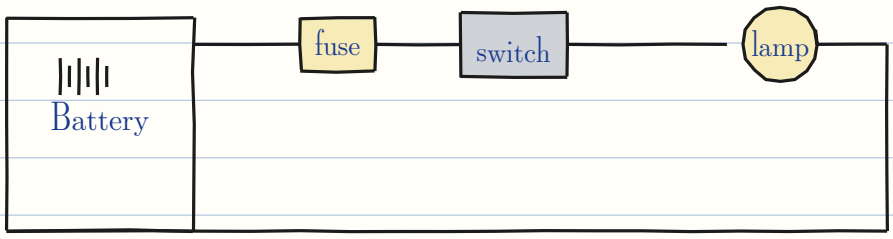


Fig 5.4 Simple vehicle circuit

Relay circuit

- A small switch energises the relay coil (85-86); the armature closes contacts 30-87 to feed the load (headlamp, horn, fuel pump) from the battery directly -> thinner switch wiring, less voltage drop.

Basic laws

- Ohm: $V = I\,R$. Power: $P = V\,I$. Fuse rating ~ 125 % of the load current; wire gauge (mm^2) chosen for voltage drop < 0.5 V.

Fuse colour (blade)	Rating
Tan / brown / red	5 / 7.5 / 10 A
Blue / yellow	15 / 20 A
Clear / green	25 / 30 A

5. Charging, Starting & Lighting Circuits

Charging system

- Alternator (3-phase AC with rectifier diodes) driven by belt; voltage regulator keeps 13.8 - 14.4 V; charge indicator lamp. Output 60-150 A; field current through slip rings & brushes.

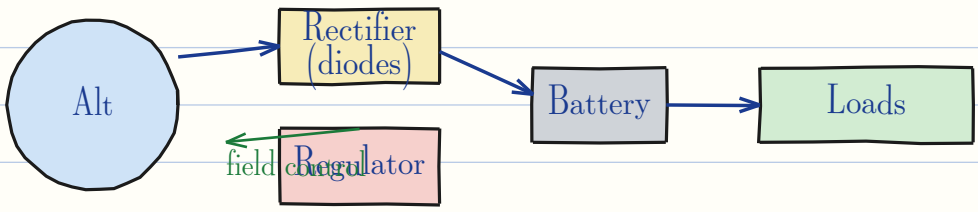


Fig 5.5 Charging circuit

Starting system

- Battery -> ignition switch -> starter solenoid (relay) -> starter motor (series-wound DC, 1-2 kW) -> pinion engages the flywheel ring gear (ratio ~ 15:1) via Bendix / pre-engaged drive; overrunning clutch protects armature.
- Cranking speed 100-200 rpm; cable thick (low resistance); neutral-safety switch for AT.

Lighting & auxiliary circuits

- Head lamps (halogen, LED, HID/xenon) high/low beam with dip switch & relay; side / tail lamps; stop lamps (brake switch); indicators with flasher unit (60-120 flashes/min); fog lamps; reverse lamp.
- Auxiliaries: horn, wipers, power windows, fuel gauge, temperature gauge, instrument cluster, central locking, CAN bus networking ECUs (up to 100 per vehicle).

Starter circuit values

Cranking current 150-300 A (petrol), up to 600 A (diesel); voltage must stay above ~ 9.6 V while cranking. Cable resistance must be below ~ 0.002 ohm.

6. Active Suspension System (ASS)

- A passive suspension has fixed spring & damper values - a compromise between comfort and handling. An active suspension uses sensors, ECU and actuators to change forces in real time -> keeps the body level and the tyres in contact on any road.

Components

- Sensors: wheel / body accelerometers, ride-height (displacement), steering angle, vehicle speed, brake pressure, yaw rate, camera road scan.
- ECU (control algorithm - sky-hook / PID / fuzzy); actuators: hydraulic / electromagnetic / pneumatic at each wheel in series with (or replacing) the spring; pump & accumulator; servo valves.

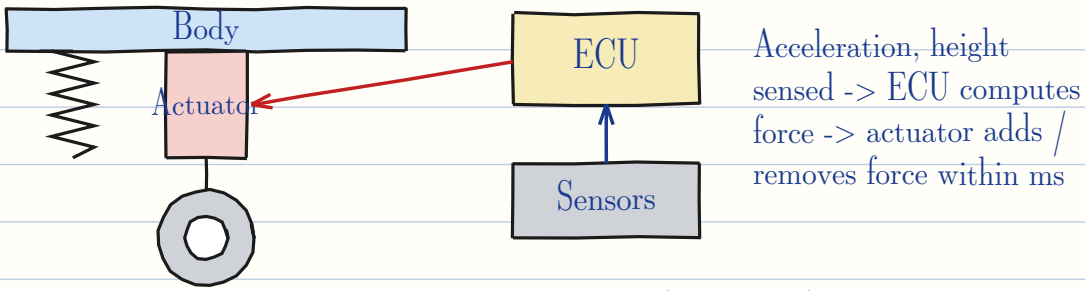


Fig 5.6 Active suspension (one wheel)

Working

- When the car hits a bump, sensors detect body acceleration; ECU commands the actuator to extend / retract and absorb it. In cornering, braking & acceleration it counteracts roll, dive and squat (body stays flat).

Advantages / disadvantages

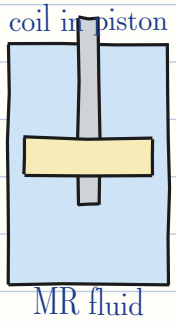
- + Excellent ride, no roll / pitch, self-levelling, adjustable height & firmness, better traction. - Costly, complex, power consumption (~ 5 kW hydraulic), maintenance.
- Examples: Mercedes Active Body Control (ABC) / Magic Body Control, Lotus active, Citroen Hydractive, Bose electromagnetic.

7. Semi-Active Suspension & MR Dampers

- Semi-active suspension: springs are passive but the damping force is varied electronically in milliseconds (no big power needs) - cheaper than fully active.

Types of control

- Electro-hydraulic variable dampers (solenoid valve changes orifice area) - VW DCC, BMW EDC, Bosch CDC.
- Magnetorheological (MR) dampers: fluid with iron particles thickens in a magnetic field -> damping changes in ~ 1 ms (Audi Magnetic Ride, Ferrari, Cadillac MagneRide).



No current: particles free -> fluid thin -> soft damping

Current ON: particles align into chains -> thick fluid -> firm damping

Fig 5.7 Magnetorheological damper

Air + adaptive systems

- Air spring + CDC (Audi adaptive air suspension; Range Rover): ride height changes (access, off-road, high-speed lowering) plus adaptive damping.
- Predictive systems: camera scans road ahead (Magic Body Control) and prepares the suspension before the bump.

Type	Spring	Damper	Power use
Passive	Fixed	Fixed	None
Semi-active	Fixed	Variable	Low
Fully active	Actuator-controlled	Actuator-controlled	High

8. Electronic Brake Distribution (EBD)

- EBD (EBFD) is a software function of the ABS system that automatically varies braking force between front and rear wheels (and left/right) depending on load, deceleration and road grip - replaces the mechanical proportioning valve.

Why needed?

- During braking, weight shifts forward -> rear wheels become lightly loaded and can lock first (spin-out risk). With a load in the boot, rear axle can take more brake force. A fixed valve cannot adapt to all these conditions.

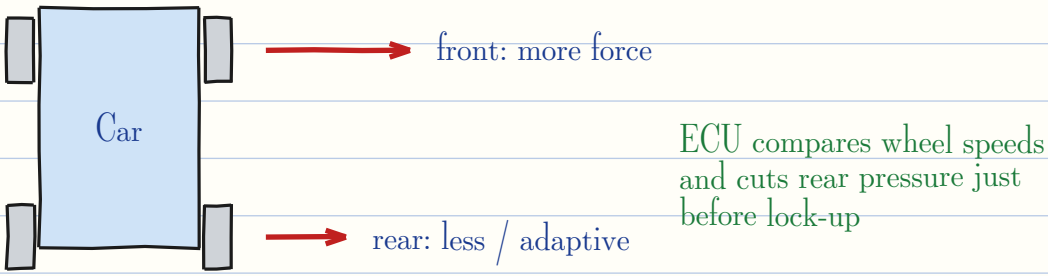


Fig 5.8 Brake force distribution under braking

Working

1. Wheel speed sensors (ABS) tell the ECU the slip of each wheel.
2. If rear wheel slip exceeds the front by a threshold, ECU closes the rear inlet valve (hold) or releases pressure (like ABS but earlier).
3. Rear brake pressure is limited, front pressure still builds to full -> shortest stop, stable car.

- Advantages: optimum use of rear grip, shorter braking distance, less pedal effort, no valve wear, adapts to load & slope, reduces brake fade at rear. Works before ABS cuts in.

9. EBD vs Proportioning Valve & Brake Assist

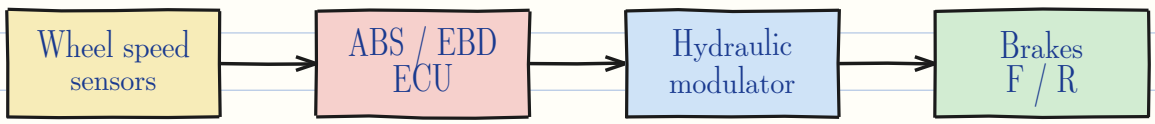
Point	Mechanical valve	EBD
Control	Fixed / load-sensing spring	Electronic, real-time
Adaptation	Little	Load, road, cornering
Cost	Low	Software on ABS hardware
Fault display	None	Warning lamp (brake / ABS)
Accuracy	Approx.	Precise

Cornering Brake Control (CBC)

- EBD extension - during braking in a bend, inside wheel carries less load; the ECU reduces pressure at the inner rear wheel to avoid oversteer / yaw.

Brake Assist (BA / EBA)

- System detects emergency (panic) braking from the speed at which the pedal is pressed and applies full brake pressure immediately even if the driver does not press hard enough (average drivers use only ~ 40-50 % force). ABS then prevents lock-up.



EBD uses the same hardware as ABS - only extra software

Fig 5.9 Hardware shared by ABS, EBD and BA

Typical ABS-family hierarchy

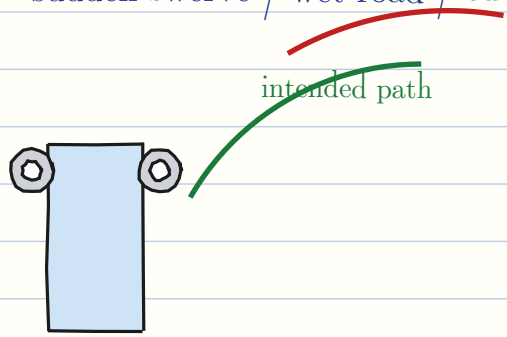
- Conventional brake -> ABS (anti-lock) -> EBD -> BA -> TCS (traction) -> ESP (stability, built on all above). A modern car has them integrated in one hydraulic control unit (HCU).
- Mandatory: ABS in all new cars in India from 2019 (also airbags), EBD with ABS.

10. Electronic Stability Program (ESP)

- ESP / ESC / DSC / VSC is an active safety system that detects and reduces loss of traction (skidding) by braking individual wheels and cutting engine torque to keep the car on the driver's intended path.

Why needed?

- Understeer - front wheels slide, car goes wider than steered (plough). Oversteer - rear slides, car spins inward. Both when the lateral grip limit is exceeded in sudden swerve / wet road / curves.



ESP brakes INNER rear wheel
in understeer -> pulls nose in

ESP brakes OUTER front wheel
in oversteer -> corrects spin

Fig 5.10 ESP corrective action

Sensors

- Steering angle (driver intent), yaw-rate (rotation about vertical axis), lateral accelerometer, wheel speed (4), brake pressure, throttle / engine torque (CAN).

Actuators

- Hydraulic unit (ABS valves + active pre-charge pump) - brakes any wheel independently, ECU requests engine torque reduction (throttle / ignition / fuel cut), warning lamp (flashes when active).

ESP = ABS + TCS + yaw control

- ESP is built on top of ABS and TCS - it adds yaw-rate & steering sensors and control software; it can brake any one wheel even when the driver is not braking.

Names for ESP

ESP (Bosch / Mercedes), ESC (generic), DSC (BMW, Jaguar), VSC (Toyota), VDC (Nissan, Subaru), StabiliTrak (GM).

11. ESP - Working & Benefits

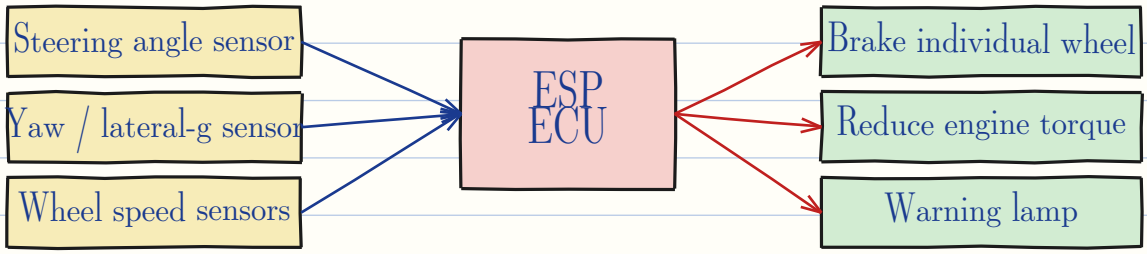


Fig 5.11 ESP block diagram

Working

1. Driver intention (steering angle + speed) is converted to a target yaw rate.
2. ECU compares it with the measured yaw rate / lateral acceleration; difference = instability.
3. Corrective yaw moment is created by braking selected wheel(s) (up to ~ 100 bar from the pump) and reducing engine torque.
4. Once vehicle follows path, intervention stops (< 150 ms reaction). Driver feels pulsation, hears pump noise, lamp flashes.

Situation	ESP action
Understeer (left turn)	Brake left rear wheel + cut torque
Oversteer (left turn)	Brake right front wheel + cut torque
Sudden lane change	Brake individual wheels in sequence
Rollover risk (SUV)	Roll-over mitigation (RMI) - brake outer front

Benefits

- Reduces single-vehicle crashes by ~ 30-40 %; mandatory in EU (2014), US (2012) and India for several classes; works on any road; off switch for off-road / snow chains.

12. Traction Control System (TCS)

- TCS (ASR, TRC, ASC) prevents the drive wheels from spinning on slippery surfaces during acceleration by limiting wheel slip -> maximum traction + stability. It is the "ABS for acceleration".

Components

- Wheel-speed sensors (shared with ABS), ECU (integrated with ABS / ESP), hydraulic modulator with pump, electronic throttle / engine ECU interface, switch & dash lamp.

Working

1. ECU compares driven wheel speed with non-driven wheel (vehicle) speed; slip > ~ 10-15 % = wheelspin.
2. Brake intervention: brake applied to the spinning wheel -> the differential transfers torque to the wheel with better grip (acts as an electronic limited-slip differential).
3. Engine intervention: reduce throttle opening, retard ignition, cut fuel injection to cylinders, ECU torque request.

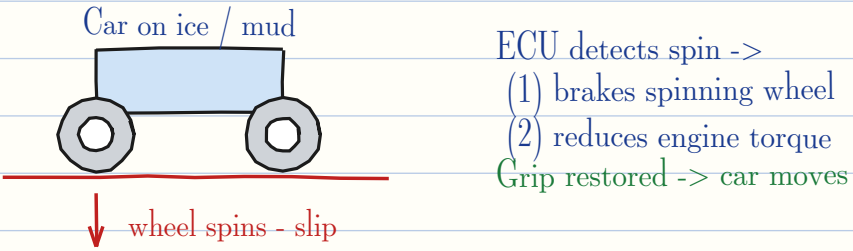


Fig 5.12 Traction control action

System	Prevents	Operates during
ABS	Wheel lock	Braking
TCS	Wheel spin	Acceleration
ESP	Skid / yaw	Cornering / any

13. Global Positioning System (GPS)

- GPS is a satellite-based radio-navigation system (USA, NAVSTAR) giving 3-D position, velocity and time anywhere on Earth, in all weather. Other GNSS: GLONASS, Galileo, BeiDou, IRNSS / NavIC (India).

Segments

- Space segment: 24+ satellites in 6 orbital planes, altitude 20,200 km, period ~ 12 h. Control segment: ground stations (master control Colorado) update clocks & orbits. User segment: receivers (car, phone).

Principle - trilateration

- Each satellite transmits a signal with its position and exact time (atomic clock). Receiver measures travel time dt -> distance d = c x dt (c = 3 x 10^8 m/s). Distances from 3 satellites fix a point on Earth; the 4th satellite corrects the receiver clock error (x, y, z, t = 4 unknowns).

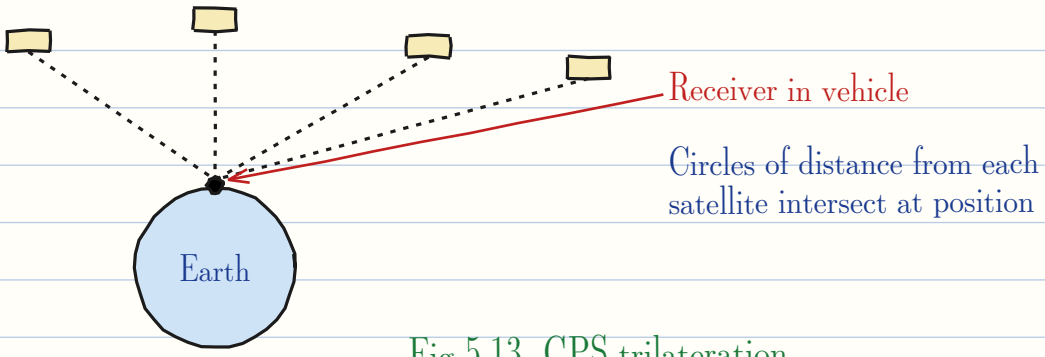


Fig 5.13 GPS trilateration

- Signals: L1 = 1575.42 MHz, L2 = 1227.6 MHz; C/A code (civil) & P code (military). Accuracy: ~ 3-5 m (civil), < 1 m with DGPS / RTK.

Key facts

- Minimum 4 satellites needed for a 3-D fix; GPS time is kept by atomic (Cs / Rb) clocks; relativity corrections are built into the signal.
- Position fix time: cold start 30-60 s, hot start 1-5 s; update rate 1-10 Hz in car units.

Remember

d = c x dt; 3 satellites -> position, 4th satellite -> receiver clock correction; accuracy ~ 5 m (civil).

14. GPS in Automobiles

Vehicle navigation system

- Components: GPS receiver + antenna, map database (digital map, SD / cloud), dead-reckoning sensors (gyro, odometer / wheel speed) for tunnels, processor + display + speaker, user interface (voice), map-matching software, traffic info (TMC / data).

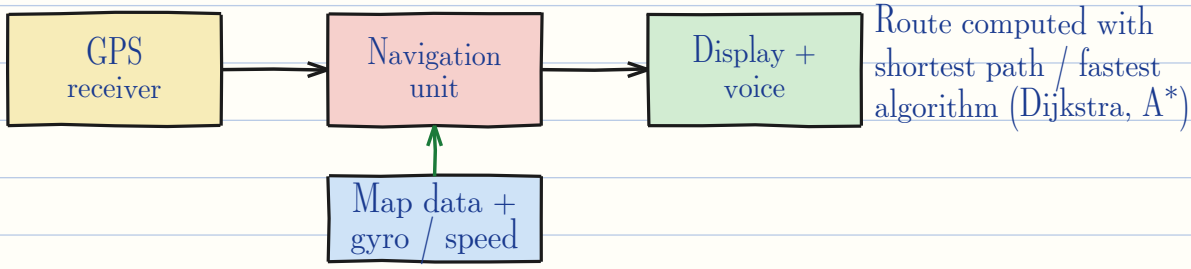


Fig 5.14 Car navigation system

Applications in automobiles

- Turn-by-turn navigation, real-time traffic & re-routing; fleet management / vehicle tracking (AIS-140 mandatory in public vehicles in India), anti-theft recovery, emergency call (eCall / SOS), geo-fencing, speed limiting, ride-hailing & logistics (Ola, Uber), insurance telematics (pay-as-you-drive), stolen-vehicle immobilisation, ADAS & autonomous driving (with lidar, HD maps).

Advantages / limitations

- + Saves time & fuel, improves safety & security, data for road planning. - Signal loss in tunnels / urban canyons, privacy concerns, multi-path errors, jamming / spoofing.
- Error sources: ionospheric delay, clock error, multipath, satellite geometry (DOP). Improved by WAAS / EGNOS / GAGAN (India) augmentation.

NavIC (IRNSS)

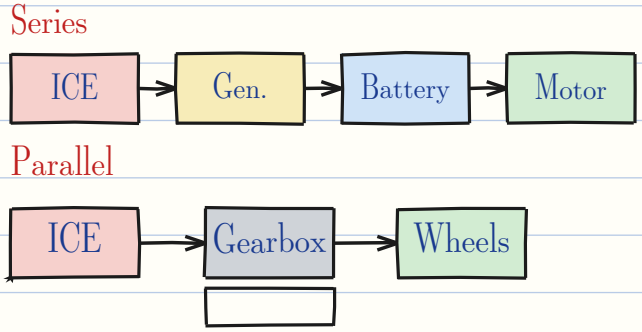
India's regional system: 7 satellites, accuracy ~ 5 m over India + 1500 km around; being fitted in vehicle tracking devices & phones.

15. Hybrid Electric Vehicle (HEV)

- A hybrid vehicle combines an internal combustion engine (ICE) with an electric motor / generator + battery to reduce fuel consumption & emissions while keeping range & performance.

Classification by architecture

- Series hybrid: ICE drives only a generator; electric motor alone drives the wheels (range-extender, BMW i3 REx, Nissan e-Power). Engine runs at best-efficiency speed.
- Parallel hybrid: both ICE & motor can drive the wheels through a common transmission (Honda IMA, mild hybrids).
- Series-parallel (power-split): planetary gear splits power; best of both (Toyota Prius, Hyryder, Hycross).



Series: engine -> electricity -> motor drives wheels

Parallel: engine and motor both push the same shaft

Power-split: Prius planetary gear

Fig 5.15 Hybrid configurations

Classification by degree of hybridisation

Type	Motor	Function	Example
Micro	2-5 kW	Start-stop only	Maruti SHVS
Mild (MHEV)	10-20 kW	Assist + regen, no EV mode	Brezza, Ciaz SHVS
Full (HEV)	30-80 kW	Electric-only driving short	Prius, Camry, City e:HEV
Plug-in (PHEV)	50-100 kW	30-80 km EV range; grid charge	Prius Prime, BYD

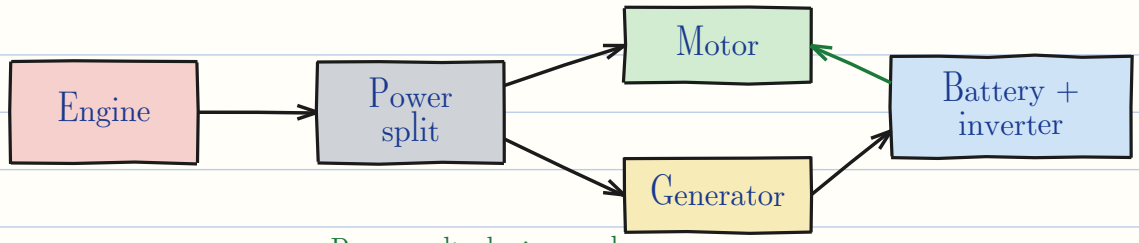
16. Hybrid - Components & Working Modes

Main components

- ICE (Atkinson cycle, high efficiency), traction motor / MG (permanent magnet AC), generator, power control unit (inverter, DC-DC converter), HV battery (Ni-MH / Li-ion 200-300 V), power-split device / CVT, hybrid ECU, regenerative braking.

Operating modes (Toyota Prius type)

Mode	Power flow
Starting / low speed	Motor only (EV), engine OFF
Normal driving	Engine + motor share, engine at efficient point
Full throttle	Engine + motor + battery boost
Deceleration / braking	Regenerative braking: motor works as generator -> charges battery
Stop	Engine OFF (idle stop)
Battery low	Engine drives generator -> charges



Power-split device = planetary gear set

Fig 5.16 Power-split hybrid drive

Advantages

- 20-40 % better fuel economy, lower CO2 / NOx, regenerative braking (recovers 15-25 % of energy), quiet city driving, no range anxiety, no external charging (HEV).

Disadvantages

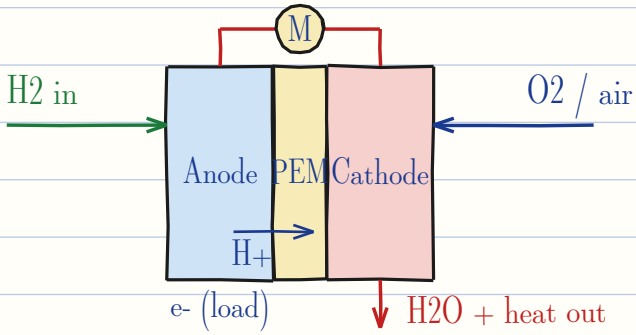
- Higher cost, extra weight, battery life / disposal, complex service & high-voltage safety (orange cables).

17. Fuel Cell Vehicle (FCEV)

- A fuel cell is an electrochemical cell that converts the chemical energy of hydrogen + oxygen directly into electricity (water + heat only) as long as fuel is supplied. Efficiency 40-60 % (vs ICE 25-35 %).

PEM fuel cell (most used in cars)

- Anode (-): H2 -> 2H+ + 2e- (platinum catalyst). Cathode (+): 1/2 O2 + 2H+ + 2e- -> H2O. Overall: H2 + 1/2 O2 -> H2O + electricity + heat. Protons pass through the Proton Exchange Membrane (Nafion); electrons flow in the external circuit -> motor.



Stack of cells in series gives 200-300 V; each cell ~ 0.7 V

H2 stored at 700 bar in carbon-fibre tank (5 kg, 500 km)

Fig 5.17 PEM fuel cell

Type	Electrolyte	Temp	Use
PEMFC	Polymer membrane	60-80 C	Cars, buses (Mirai, Nexo)
AFC	KOH	90-100 C	Space
PAFC	Phosphoric acid	200 C	Stationary
SOFC	Ceramic	800-1000 C	Power plants, APU

- Advantages: zero tailpipe emission, quiet, quick refuelling (3-5 min), long range. Challenges: H2 production cost / storage, platinum cost, infrastructure, durability.